# The Puzzle of Partial Retirement and Liquidity Constraints*

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**Abstract**

The life-cycle hypothesis predicts that older workers will use partial retirement to transition gradually toward full retirement, yet observed participation remains low. While the literature often attributes abrupt retirement to employer-side constraints, we propose a different mechanism: liquidity constraints. Drawing on administrative data from the largest pension fund in the Netherlands, we study partial retirement behavior using two major pension reforms and a dynamic life-cycle model. We exploit one reform for estimation and the other for out-of-sample validation, ensuring credible policy simulations. In line with the life-cycle hypothesis, we show that older workers who can smooth consumption over partial retirement years are those who can afford to retire gradually. High or subsidized part-time wages, generous pension provisions, or personal savings allow workers to bridge the income gap and accommodate preferences for a gradual exit. We demonstrate that policies facilitating consumption smoothing during partial retirement make gradual transitions attractive.

## 1 Introduction

In the standard life-cycle model, individuals choose consumption and leisure to maximize lifetime utility. Because of the diminishing marginal utility of leisure, the model predicts a gradual reduction in hours worked as individuals approach retirement. In this framework, the absence of liquidity constraints allows workers to supplement part-time earnings by drawing down accumulated savings, maintaining a steady consumption profile while transitioning out of the workforce. This gradual transition arises naturally in a labor-supply problem with a convex budget set and concave preferences, which guarantees that intermediate hours are feasible and that optimal labor supply adjusts smoothly (Gustman and Steinmeier, 1986). Consistent with

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this theoretical preference for smoothing, stated-preference data suggests that more than one in three individuals would prefer partial retirement over an abrupt exit (Kantarcı et al., 2025).

Despite these preferences, the gradual reduction of hours predicted by standard theory is rarely observed. Rogerson and Wallenius (2013) demonstrate that abrupt, or 'cliff-edge', retirement arises as an optimal response to production nonconvexities, such as fixed costs or part-time wage penalties. Reconciling such transitions within a standard convex model requires a leisure elasticity of intertemporal substitution (EIS) of 0.75 or higher, far exceeding typical micro-estimates of approximately 0.25. This implies that an EIS estimate depends critically on the assumed strength of nonconvexities. Building on this insight, Ameriks et al. (2020) use stated-preference data to recover older workers' labor EIS in an environment that is not constrained by demand-side frictions. Their estimates indicate a modest elasticity consistent with micro-evidence, suggesting that abrupt exits are not a result of high leisure substitutability but are instead driven by demand-side constraints.

In this paper, we argue that the prevailing focus on demand-side constraints overlooks a fundamental supply-side mechanism: the worker's inability to bridge the income gap resulting from partial retirement. While the literature often attributes abrupt retirement to firm-level frictions or production nonconvexities, we demonstrate that the transition is limited by liquidity constraints within the worker's own lifecycle budget set. Because partial retirement entails a substantial reduction in labor income, insufficient liquidity can push workers toward a cliff-edge exit, even when preferences favor a gradual transition. By isolating this mechanism, we show that liquidity constraints can substantially limit gradual retirement, even in the absence of strong technological or demand-side restrictions.

Existing research provides suggestive evidence on the role of consumption smoothing. Johnson (2011) argues that many workers in the US can afford to reduce their work schedules only if they can access retirement benefits. This is often difficult because federal tax rules prohibit defined-benefit plans from paying benefits to active employees under age 62. Furthermore, defined-benefit formulas typically tie pension payments to earnings near the end of the career; workers who reduce their hours before retiring therefore risk losing substantial pension wealth. Other benefit systems, including health, life, and disability insurance, are similarly built around full-time eligibility rules, making it difficult for employers to extend these benefits to part-time phased retirees. Kantarcı et al. (2025) show that subsidizing part-time wages during partial retirement makes partial retirement much more attractive. Using stated-preference data from the Netherlands, they find that if part-time wages were subsidized by 10 percentage points at age 63, the probability of choosing to work 60% of former full-time hours until the statutory retirement age of 67 would increase by 6.6 percentage points.

While these studies highlight the importance of consumption smoothing, testing the underlying liquidity mechanism requires an empirical setting where access to wealth shifts independently of financial work incentives. In many pension systems, the liquidity shock of becoming eligible for benefits occurs simultaneously with a price shock to leisure, such as a tax on earnings or a budget-set kink that penalizes continued employment. For example, the widely studied Social Security Earnings Test creates such a structure, confounding the effect of gaining access to benefits with the penalty on working and often inducing abrupt retirement (French, 2005). To isolate the role of liquidity, one requires an environment where the marginal payoff to working remains undistorted upon gaining access to pension wealth. The Dutch institutional environment provides precisely this structure: the mandatory occupational pension system allows for simultaneous receipt of pensions and wages without distorting work incentives. In addition, employees have a statutory right to request reduced working hours, facilitating gradual retirement. This institutional decoupling enables us to study the role of liquidity in retirement behavior and assess whether constraints on the worker side, rather than technological or demand-side nonconvexities, limit gradual retirement.

Within this unique institutional setting, two major pension reforms, implemented in 2006 and 2013, help to examine the role of pension liquidity alongside wealth and price incentives. The 2006 reform abolished a generous, subsidized early-retirement (VUT) scheme and replaced it with an actuarially neutral system. This policy generated a negative lifetime wealth effect by reducing expected pension wealth, and a positive price effect by eliminating the implicit tax on remaining employed past age 60. It also generated a liquidity constraint by cutting off early cash-flow access. The 2013 reform gradually raised the statutory retirement age (SRA), delaying access to the state pension while retaining the accrual rate and actuarial neutrality of the occupational pension. For individuals retiring prior to the reformed SRA, this policy generated a negative lifetime wealth effect, and strengthened the price effect to remain employed past the pre-reform retirement age. It also tightened the liquidity constraint by delaying unpenalized income access.

In a standard life-cycle model without borrowing constraints, forward-looking workers absorb a wealth shock by smoothing leisure continuously across all remaining working years, ruling out sudden drops in hours at specific age thresholds. Similarly, the price incentives in our setting lack the structural non-linearities required to generate discontinuous adjustments: the 2006 reform removed a structural implicit tax rather than introducing a new one, while the 2013 reform maintained actuarially neutral accrual around the new eligibility age. Consequently, discontinuous labor-supply adjustments occurring precisely when pension cash access shifts point directly to borrowing constraints within the life-cycle framework. Specifically, by terminating early subsidized pensions, the 2006 reform removed the liquid wealth buffer required to finance early workforce exits or offset earnings losses from gradual retirement. While generating a negative wealth effect for early claimers, the 2013 reform shifted the timing of pension liquidity, moving the precise threshold at which workers can use pension cash as a bridge to support a gradual transition out of full-time employment.

Besides supply-side liquidity constraints, demand-side employer constraints alone cannot explain the lack of gradual retirement. If firm-level frictions were the primary barrier, workers would be unable to execute discrete labor-supply reductions precisely at pension eligibility thresholds.

To evaluate these theoretical predictions, we estimate the effect of the pension reforms on discrete labor-supply adjustments around the policy thresholds using a difference-in-differences (DiD) design. Focusing on the intensive margin, we show that the 2006 reform generated, for example at age 60, an abrupt 2.73 percentage point reduction in part-time work for the cohort for whom early pension liquidity was abolished. Focusing on the intensive margin, we show that the 2006 reform sharply reduced part-time work for the affected cohort, by 2.73 percentage points at age 60, for example, following the abolition of early pension liquidity. Conversely, the 2013 reform led to a 6.92 percentage point increase in part-time work at age 65 for the cohort whose statutory retirement age was delayed, as workers utilized partial pensions as a bridge upon reaching cash eligibility.

While these reduced-form estimates strongly point to liquidity constraints, a reduced-form design alone cannot fully isolate liquidity from co-occurring wealth and price effects, nor can it isolate the exact magnitude of the liquidity effect on labor supply. To disentangle these underlying mechanisms, we develop a dynamic structural life-cycle model. In the model, forward-looking agents choose consumption continuously and make discrete labor-supply decisions jointly with (partial) pension-claiming choices. We integrate the full complexity of the Dutch institutional environment, including non-linear taxation and actuarial pension rules, to precisely characterize the liquidity constraints shaping transitions at both the extensive and intensive margins. We estimate the model using rich administrative data on hours, wages, disability benefits, and assets. We also observe accrued pension rights, which avoids the measurement error inherent in reconstructed earnings histories. By observing changes in all variables at exact calendar dates, we capture high-frequency transitions often masked in coarser annual or biennial panels in the literature.

We estimate a consumption EIS of 0.73 and a leisure EIS of 0.94. The consumption EIS implies a moderate desire to smooth consumption over time and is consistent with estimates from micro studies of intertemporal consumption behavior (Blundell et al., 2008; Attanasio and Weber, 2010; Meghir and Pistaferri, 2011; Aguiar and Hurst, 2013). The leisure EIS indicates a high willingness to shift leisure across periods. Although this estimate exceeds typical micro-econometric Frisch elasticities (e.g., Chetty et al., 2011), it aligns with the larger intertemporal labor-supply elasticities often found in the macroeconomic literature (Keane, 2011; Keane and Rogerson, 2012). This preference structure implies that individuals are sensitive to liquidity constraints: they retire abruptly upon pension eligibility because pension income can replace labor earnings at a level that maintains a stable consumption path. Without these liquidity constraints, the consumption-smoothing motive would favor a more gradual transition into retirement.

Using the estimated life-cycle model, we conduct counterfactual simulations to isolate the role of pension illiquidity in constraining gradual retirement. First, we consider a scenario in which individuals can claim partial pension benefits while working part-time with actuarially fair adjustments, compared to a baseline where this option is unavailable. Because this policy alters only the liquidity and timing of pension income while remaining neutral with respect to lifetime wealth and marginal price incentives, it directly tests whether preferences accommodate gradual retirement once liquidity constraints are relaxed. Expanding the set of feasible intertemporal allocations in this way increases participation in partial retirement by 2.04 percentage points at age 63. When we instead subsidize part-time wages by 10 percentage points, increasing disposable income during partial retirement, participation rises by 4.19 percentage points at age 63. Notably, however, a subsidy to pension accrual during partial retirement, which raises income only during full retirement, generates a near-zero response. Taken together, these counterfactuals show that policies providing liquidity when earnings fall induce the gradual retirement behavior implied by standard life-cycle preferences.

We repeat these simulations across three birth cohorts exposed to different pension rules under the pension reforms studied. This allows us to examine how labor-supply responses to our policy experiments vary with the underlying baseline incentives created by the reforms. This cross-cohort heterogeneity provides further evidence that liquidity constraints drive partial retirement behavior. For cohorts facing a higher SRA, eligibility for partial pension access relaxes a prolonged period of pension illiquidity. Consequently, granting partial pension access at age 63 increases participation in partial retirement by an additional 1.31 percentage points for the cohort whose SRA was increased by 13 months. We find qualitatively similar cohort differences when subsidizing part-time wages, reinforcing our conclusion that responsiveness to partial-retirement policies is primarily driven by the need to bridge income gaps. Finally, we show that despite these sizable shifts into partial retirement, the net effect on overall lifetime labor supply remains modest.

These findings contribute to several strands of the literature. First, we add to the small body of research that structurally estimates the leisure EIS using micro data. While much of the existing literature relies on calibrated macro models to set this parameter (e.g., Prescott et al., 2009; Rogerson and Wallenius, 2013), we provide direct micro-econometric estimates recovered from observed behavior. In standard macro models, explaining abrupt retirement requires a high labor EIS (0.75 or higher) unless production nonconvexities or demand-side restrictions are introduced (Rogerson and Wallenius, 2013; Ameriks et al., 2020). Although the labor EIS and leisure EIS are mathematically distinct and cannot be directly compared due to the hours-weighting of non-market time, models in both fields typically rely on technological or demand-side barriers to rationalize abrupt transitions. We offer a new perspective that bypasses this reliance on nonconvexities by providing an alternative supply-side financial mechanism. By estimating a high leisure EIS (0.94), we demonstrate that individuals possess a high structural flexibility regarding when they consume leisure time. Rather than implying that workers prefer abrupt retirement, this high elasticity suggests they would eagerly smooth their transition if given the opportunity. Our structural framework indicates that the observed prevalence of cliff-edge retirement is not a reflection of an inherent distaste for gradual retirement, but is instead driven by binding liquidity constraints that render a phased transition financially infeasible.

Second, we contribute to the literature on partial retirement. Two macroeconomic questions about aggregate retirement behavior dominate recent research. The first concerns the prevalence of abrupt retirement, as relatively few workers reduce hours gradually. While the leading explanation emphasizes employer-side constraints (Hutchens, 2010; Rogerson and Wallenius, 2013; Ameriks et al., 2020), we argue instead that liquidity constraints make partial retirement unattractive. We show that supplementing part-time earnings with partial pensions, or directly subsidizing part-time wages, relaxes these constraints and enables individuals to smooth consumption. By contrast, subsidizing pension accruals during partial retirement has limited impact because it does not provide liquidity when workers need it most.

The second macroeconomic question concerns the total labor-supply effect of partial retirement. Incentives for partial retirement can reduce total labor supply if the availability of part-time work arrangements displaces full-time employment more than it delays full retirement. A growing literature analyzes this trade-off. Huber et al. (2016) and Berg et al. (2020) exploit quasi-exogenous variation in the timing of firms' and collective-bargaining agreements' adoption of Germany's statutory partial-retirement program and find positive total labor-supply effects. Börsch-Supan et al. (2018) exploit cross-country variation in reforms and find no evidence that flexible pension schemes increase total labor supply. Taking structural approaches, Ameriks et al. (2020) and Kantarcı et al. (2025) combine stated-preference data from hypothetical job and retirement scenarios, respectively, with life-cycle models and simulate positive effects of increased retirement flexibility on late-career labor supply. We advance this literature by integrating a complete fiscal and pension system into a dynamic framework validated against administrative data. Evaluating partial pension access, wage subsidies, and accrual subsidies as three key reforms, we find a negligible net impact on total labor supply. This result stems from two opposing behavioral responses: while the policies successfully relax liquidity constraints and induce workers to delay full exit, they simultaneously enable full-time workers to act on their high leisure elasticity and transition into part-time work earlier. Our findings therefore suggest that providing liquidity during the retirement transition facilitates a more preferred allocation of late-life leisure without significantly expanding the total volume of labor hours.

We also contribute to the broader literature on labor-supply responses to retirement incentives. This literature has traditionally focused on extensive-margin adjustments such as labor-force exit and retirement timing; see Feldstein and Liebman (2002) and Coile (2015) for reviews. Leading studies on extensive-margin responses include French (2005), Coile and Gruber (2007), Manoli and Weber (2016), Fetter and Lockwood (2018), and Seibold (2021). More recent structural work that incorporates both extensive- and intensive-margin choices includes Fan et al. (2024), Iskhakov and Keane (2021), and Artmann et al. (2026). We document sizable behavioral adjustments at both the extensive and intensive margins, building on a smaller body of work examining hours adjustments among older workers (Gustman and Steinmeier, 1986, 2000, 2004; Berkovec and Stern, 1991; Gelber et al., 2020). Methodologically, we pair workhorse treatment-effect designs that identify the causal impacts of pension reforms with a structural life-cycle model that simulates counterfactual policy reforms. Substantively, we depart from earlier studies by uncovering a mechanism through which retirement incentives shape labor-market behavior: generous pension provisions relax liquidity constraints, whereas delayed access to pension wealth tightens them. While these policy shifts change lifetime income, individuals who can freely borrow or save would spread this income shock smoothly across their entire life cycle. Such unconstrained wealth effects would predict a gradual adjustment in hours rather than sharp behavioral changes precisely at eligibility ages. In contrast, we demonstrate that both forces induce abrupt, age-specific transitions into partial retirement, highlighting the primary role of liquidity constraints over pure lifetime wealth effects.

We contribute to the structural modeling literature in several respects. First, we follow the growing body of research that integrates (quasi-)experimental variation into structural estimation (Low and Meghir, 2017; Andrews et al., 2020). While existing studies typically use such variation either to aid estimation (French et al., 2022; Low and Pistaferri, 2015; Autor et al., 2019; Best et al., 2020; DellaVigna et al., 2017) or to validate model predictions (Todd and Wolpin, 2006; Attanasio et al., 2012; De Bresser, 2024), we exploit two distinct pension reforms to combine both approaches. We use one reform for estimation and the other for out-of-sample validation, and then reverse these roles to assess robustness. Our consumption and leisure EIS estimates remain nearly identical regardless of which reform provides identification. This stability across different policy environments suggests these are fundamental preference parameters rather than mere artifacts of a specific institutional setting, providing strong empirical support for our counterfactual simulations. Heuristically, the model identifies these parameters by matching the joint distribution of labor supply and assets in response to shifts in the "liquidity bridge". Within each reform, the leisure EIS is identified by the high sensitivity of both the extensive and intensive margins to the timing of pension eligibility, revealing a high willingness to substitute leisure over time. Simultaneously, the consumption EIS is identified by the prevalence of cliff-edge exits when supplemental income is absent, indicating that individuals are unwilling to sustain the consumption drop required for a gradual transition without immediate liquidity.

Second, we address the initial-conditions problem and the bias inherent in approximated state variables by using administrative records that observe accrued pension rights. While standard structural models must reconstruct or approximate past earnings (e.g., French, 2005; Blau and Gilleskie, 2006; French and Jones, 2011), we initialize pension wealth exactly. By eliminating the measurement error inherent in such reconstructions, we prevent the propagation of errors through the recursive structure of the model and ensure that preference parameters are disciplined by the agent's actual intertemporal budget set rather than an imputed proxy.

Third, the high-frequency nature of our administrative data allows us to identify the timing and sequencing of (partial) retirement with greater precision. We observe the exact dates of changes in all variables, avoiding the undercounting and blurring of exit paths common in coarser biennial panels (Blau, 1994). This granularity ensures that mid-year transitions are mapped to an individual's true age-year rather than being misclassified by rigid calendar snapshots. This precision provides a sharper test of forward-looking behavior: by capturing exactly how fast individuals react to age-specific incentives, we can cleanly separate planned, long-term lifecycle choices from immediate, short-run responses. This prevents timing errors from biasing our structural estimates of intertemporal preferences and adjustment frictions.

Fourth, we advance the structural modeling of late-career labor supply by endogenizing the intensive margin within a complex, nonconvex budget set. Existing structural models increasingly allow for part-time work or continuous hours, but they do not permit workers to claim pension benefits while continuing to work and accrue additional rights, a central institutional feature of the Dutch system. Leveraging this environment, we incorporate hours reductions, partial pension claiming, and continued accrual directly into the intertemporal budget set. Modeling these features jointly allows us to identify the interaction between liquidity constraints and wealth effects and to recover preference parameters that would be difficult to pin down in frameworks omitting these institutional margins.

The remainder of the paper is structured as follows. Section 2 describes the Dutch pension policy design and major reforms. Section 3 presents the pension and labor market data and the sample restrictions. Section 4 provides descriptive analyses of employment patterns and partial retirement. Section 5 presents causal evidence on the impact of pension reforms. Section 6 introduces the dynamic model of retirement. Section 7 presents MSM estimates of the dynamic model. Section 8 conducts policy simulations. Section 9 concludes.

## 2 Pension policy in the Netherlands: Institutional design and major reforms

Retirement income in the Netherlands mainly stands on two pillars: the state and the occupational pension.[1] The General Old-Age Pensions Act (AOW) is the state pension scheme, paying a flat-rate benefit when people reach the state pension age, independent of earnings, income or premiums paid. The benefit level depends on the number of years of residence in the country and on household composition. For those who always resided in the country, it provides households older than the state pension age with a subsistence-level income. The scheme is unfunded and based on the pay-as-you-go principle: current state pensions are financed from the current premiums paid by workers. The premiums are paid through income tax. The scheme does not allow flexible claiming of pension rights.

Participation in the fully funded occupational pension scheme is mandatory for the vast majority of employees. The scheme follows a defined-benefit (DB) structure and, while essentially individual, it includes provisions for surviving spouses and orphans.

From the early 1990s until 2005, Dutch workers could retire well before the statutory pension age through generous early-retirement arrangements known as VUT schemes. These provisions were occupational arrangements, established through sectoral collective labor agreements and financed by mandatory payroll contributions from all employees and employers in the sector. Workers contributed simultaneously to the funded DB occupational pension scheme and to the VUT scheme, but only the former generated individual pension entitlements. VUT operated as a separate pay-as-you-go early-retirement insurance layer: contributions from active workers financed the benefits of older workers, without accumulating individual pension wealth or actuarially fair rights. Eligibility typically required long tenure within the sector — often 10 to 20 years — and reaching the sector-specific VUT age, which in some public-sector agreements was as low as 55. Benefits were highly generous, commonly around 70% of the last gross wage, and claiming VUT did not reduce the occupational pension later received under the regular funded DB scheme. Because VUT rights were embedded in collective agreements, widely communicated by unions and employers, and routinely used by colleagues, workers were well aware of their eligibility. As a result, early retirement through VUT became a socially normalized pathway for workers with stable employment histories.

The Netherlands implemented two major pension reforms in the last two decades. Generous early-retirement arrangements were gradually phased out after a 2006 tax reform (RVU) made employer-financed early-retirement benefits financially unattractive. For workers who were still

[1] Knoef et al. (2016) show that mandatory pension savings constitute 72%, and the present value of housing wealth constitutes 12.6% of all income available at age 65. The remaining fraction is private savings meaning that they are of little importance to retirement income.

in the labor force when the 2006 reform took effect, a transitional arrangement preserved access to the FPU early-retirement option. Eligible individuals were allowed to retain the right to claim their occupational pension at age 62 without an actuarial reduction. By eliminating the generous early pension, the 2006 reform reduced expected pension wealth, generating a negative lifetime wealth effect. Simultaneously, because remaining employed no longer meant losing out on these early pension payments, the reform removed the implicit tax on work past age 60, generating a positive price incentive. By restricting access to unpenalized pension benefits prior to age 62, the reform also introduced a liquidity constraint on early exit.

After the 2006 reform, occupational pension funds integrated early, late, and partial retirement into the regular DB scheme on an actuarially neutral basis. Although early and partial retirement were already possible in some pension funds before 2006, the disappearance of subsidized VUT pathways made actuarially neutral retirement — typically between ages 60 and 70 — the dominant framework.

The SRA, the age at which individuals become entitled to the state pension, was fixed at 65 for many years, applying to all birth cohorts reaching age 65 up to and including 2013. In 2011, the government announced a gradual increase in the SRA. For example, individuals born between March 1, 1957, and December 31, 1959, became eligible at age 67, while those born in 1962 faced an eligibility age of 67 years and 3 months. Based on current projections, the state pension age will continue to rise in line with life expectancy, reaching 68 by 2040 and 70 by 2070. The benefit level of the state pension remained unchanged, and the occupational pension retained both its accrual rate and actuarial neutrality. For individuals retiring prior to the reformed SRA, this policy generated a negative lifetime wealth effect by losing the state pension and receiving a lower occupational pension annuity, and it strengthened the price incentive to remain employed, as continuing to work avoids these actuarial penalties and builds additional pension wealth. Importantly, it also tightened the liquidity constraint by delaying access to unpenalized pension cash flows.

As a result of the phasing out of generous early retirement schemes and the rising SRA, the average retirement age increased from about 61 in the early 2000s to nearly 65 by the late 2010s, and continues to rise, reaching around 66.5 by the mid-2020s. These reforms have made early retirement increasingly difficult for older workers, particularly those with health issues or physically demanding jobs. In response, employer and employee organizations introduced subsidized partial retirement schemes ("Generation pact") through collective labor agreements in the late 2010s. Typically starting at age 60 or 62 depending on sector, these allow employees to reduce working hours in the years leading up to their SRA, with a less-than-proportional drop in income and continued pension accrual based on full-time earnings. Sector-specific details vary; a common example is the 80/90/100 variant, where employees work 20% less, earn 90% of their full-time wage, and accumulate pension rights as if working full-time.

Beyond these specific retirement schemes, the broader legal environment provides a foundational right for employees to adjust their labor supply. Under the Working Hours Adjustment Act (Wet aanpassing arbeidsduur) since 2000, and its 2016 successor, the Flexible Working Act (Wet flexibel werken), employers are legally required to honor employee requests for reduced hours unless they can demonstrate compelling business interests for refusal.

To exploit differences in retirement incentives across policy regimes, we analyze six birth cohorts. Table 1 presents these cohorts, detailing their ages at the start and end of the dataset's observation period, the earliest age at which they can claim occupational pensions, and the SRAs at which they can claim state pensions. The first and second cohorts qualify for the state pension at the same age; however, only the first cohort is eligible for generous occupational pensions starting at age 55. In contrast, the second cohort must wait until age 60 to claim occupational pensions. The younger cohorts face progressively higher statutory pension ages, although all retain the option to claim occupational pensions from age 60.

The subsidized partial retirement schemes are unlikely to have affected the observed birth cohorts for two reasons. First, they were introduced only in the late 2010s, near the end of our observation period. Second, by that time, many individuals in these cohorts were already past the typical eligibility ages, often around 60 or 62, for entering such schemes. We therefore do not consider these subsidized schemes as relevant retirement incentives for the cohorts analyzed.

To illustrate the financial consequences of the 2006 reform, consider two otherwise identical workers who both stop working at age 60. Under the pre-reform system, early retirement at age 60 yielded an unpenalized pension equal to 70% of the last gross wage. For a worker earning €40,000, this corresponds to an annual pension of €28,000. Under the reformed actuarially neutral DB scheme, the same worker receives the full 70% replacement rate only when retiring at age 65. Retiring five years earlier leads to an actuarial reduction of about 6% per year, implying a 30% cut relative to the age-65 benefit. The replacement rate at age 60 therefore falls to 49%, or €19,600 per year. At age 60, the reform lowers annual pension income by €8,400, reflecting a negative lifetime wealth shock and introducing an immediate liquidity constraint on early exit.

A similar back-of-the-envelope calculation illustrates the financial consequences of the increase in the SRA. Consider two identical workers who both stop working at age 65 years and 3 months. For the older cohorts in Table 1, the SRA is 65 years and 3 months, so the worker immediately receives the full state pension together with the actuarially neutral occupational pension. Assuming a representative annual state pension of €15,000 and an occupational pension of €20,000, total pension income at age 65 years and 3 months amounts to €35,000 per year. Younger cohorts, however, face higher SRAs, such as 66 years and 4 months for the 1953 cohort. Therefore, a worker who stops working at 65 years and 3 months receives only the actuarially reduced occupational pension until reaching the higher SRA, while the state pension is postponed accordingly. Using a standard actuarial reduction of 6% per year, retiring 1 year and 1 month before the neutral age lowers the occupational pension by 6.5%, from €20,000 to €18,700 per year. Over the period until state-pension eligibility, total pension income is therefore €18,700 annually instead of €35,000, implying an annual shortfall of €16,300. In total, the SRA increase reduces pension income by about €17,660 over the 1 year and 1 month gap, unless the worker postpones retirement accordingly. While part of this gap reflects a reduction in expected lifetime wealth due to delayed state pension payments and lower annual occupational payouts, the actuarial reduction on the occupational pension also creates a price incentive to remain employed to avoid a permanent benefit cut. Crucially, this creates a severe short-term liquidity constraint during the gap.

Table 1: Birth cohorts, their ages at the start and end dates of the observation period of the dataset, and their minimum retirement ages for occupational pensions and SRAs for state pensions

| Birth cohort | Birth date | Age in years and months on 01.01.2005 (start of observation period) | Age in years and months on 12.31.2019 (end of observation period) | Minimum age of occupational pension eligibility | Statutory retirement age of state pension eligibility in years and months |
|---|---|---|---|---|---|
| 1 | 01.11.1949 – 31.12.1949 | 55+2 – 55+0 | 70+2 – 70+0 | 55 | 65+3 |
| 2 | 01.01.1950 – 30.09.1950 | 55+0 – 54+3 | 70+0 – 69+3 | 60 | 65+3 |
| 3 | 01.10.1950 – 30.06.1951 | 54+3 – 53+6 | 69+3 – 68+6 | 60 | 65+6 |
| 4 | 01.07.1951 – 31.03.1952 | 53+6 – 52+9 | 68+6 – 67+9 | 60 | 65+9 |
| 5 | 01.04.1952 – 31.12.1952 | 52+9 – 52+0 | 67+9 – 67+0 | 60 | 66+0 |
| 6 | 01.01.1953 – 31.07.1953 | 52+0 – 51+5 | 66+11 – 66+5 | 60 | 66+4 |

# 3 Administrative data and sample restrictions

ABP is the largest pension fund in the Netherlands and ranks among the largest globally, covering approximately 15% of the Dutch population. It serves as the occupational pension fund for employees working in the government and education sectors.[2] Our analysis is based on administrative records from ABP, covering all individuals who are observed participating in a pension scheme between January 2005 and December 2019. Within this time period, we observe individuals born between 1940 and 1974.

From January 2005 to December 2019, we observe the date of birth, gender, marital status, gross wages (before taxes and employee social security contributions), full-time equivalent (FTE), and accrued pension rights. In the life-cycle model, the accrued rights allow us to initialize individuals' pension wealth exactly and to project future entitlements using their full contribution history. We also observe paid pension rights starting from January 2006. FTE is defined as the ratio of actual hours worked to full-time hours, with the latter set at 38 hours per week in sectors insured by ABP. We observe the precise dates at which changes occur in these variables, avoiding the undercounting of transitions that arises in coarser data (Blau, 1994). Using this information, we construct a panel dataset with monthly observations at the individual level.

Using individuals' unique citizen service numbers (BSN), we link ABP administrative records to those from Statistics Netherlands (CBS), which provide additional information on household assets and disability insurance (DI) benefits. Asset data are available annually and include financial assets, bank savings, real estate holdings, debts, and mortgages. DI benefits are available monthly. Both datasets are observable throughout our observation period.

We impose several restrictions on the dataset. First, we restrict our sample to individuals born between November 1949 and July 1953. As Table 1 showed, this range captures career periods during which pension policies changed and individuals were most likely to make retirement decisions. We exclude individuals born before November 1949 because our analysis requires variation across birth cohorts in either the minimum age for occupational pension eligibility or the SRA for state pension eligibility, but not in both simultaneously. For cohorts born before November 1949, both eligibility ages (55 and 65, respectively) differ from those of all cohorts included in Table 1. Individuals born after July 1953 are excluded, as they reach their SRA only after the end of our observation period in December 2019. Consequently, we cannot observe retirement decisions triggered by reaching the statutory pension age for this group.

Second, we focus on individuals who are employed at the age of 55 years and 2 months, the youngest age the oldest birth cohort is observed at the start of the observation period as Table 1 showed. This selection is not restrictive since approximately 85% of the individuals in the original dataset are working at this age.

Third, we limit our analysis to men. Among employed women in the dataset, 60% work part-time, and this proportion remains stable both between ages 55 and 65 and across birth cohorts. This pattern suggests that many women work part-time throughout their careers and that their part-time choices did not adjust in response to the pension reforms, despite changes in lifetime income and liquidity constraints. Since our objective is to examine patterns of gradual retirement from full-time positions, we exclude women from our analysis.

Finally, we exclude individuals who, during the observation period, transitioned from a sector insured by ABP to one not insured by ABP, thereby accruing pension rights in a different fund. This restriction affects only a small subset of the sample and does not influence our qualitative

[2]The government sector includes the central government, provinces, municipalities, water boards, the military, police, judiciary, and all civil servants. The education sector encompasses all levels of schooling, universities, public research institutes, and university medical centres.

conclusions.

These sample restrictions yield a final analytical sample comprising 8,519,210 observations for 62,402 individuals. These individuals fall into the six birth cohorts detailed in Table 1 as follows: 2,456 in the first cohort, 11,554 in the second, 12,358 in the third, 12,562 in the fourth, 13,566 in the fifth, and 9,906 in the sixth cohort.

# 4 Employment patterns in late career and partial retirement

Figure 1 presents full- and part-time employment rates over time for individuals aged 55 and 2 months to 66 years and 11 months, the earliest and latest ages at which all birth cohorts are observed throughout the available data period. We define full-time work as working at least 0.875 FTE. As explained in Section 3, the sample is restricted to men who were employed at age 55 years and 2 months. The figure shows that approximately 90% of the sample is engaged in full-time work at this age. This high rate of full-time employment is desirable for our analyses, as it provides a large baseline population from which transitions into abrupt or partial retirement can be observed and compared across cohorts.

The proportion of individuals engaged in full-time employment declines progressively with age, with pronounced declines at age 62 for birth cohort 1, and near SRAs for all cohorts. These patterns are consistent with institutional eligibility thresholds and actuarial rules: individuals in cohort 1 become eligible to claim their occupational pension without an actuarial reduction at age 62, and all cohorts become eligible for state pensions at the SRA (Table 1).

Cohort 1 consistently shows lower rates of full-time labor market participation between ages 56 and 65, driven by higher early retirement rates or increased part-time work between ages 56 and 62, both enabled by access to VUT benefits and early occupational pension claims prior to age 62.

Cohorts 2-6 retire later than cohort 1, reflecting increases in the SRA. Compared to cohort 2, full-time employment is already more prevalent in cohort 6 beginning at age 60. This pattern is consistent with forward-looking labor-supply behavior. When the SRA increases, individuals reassess their entire late-career trajectory: they anticipate a longer period until public pension eligibility and recognize that exiting the labor force early will leave them without pension income for a longer time. In response, they adjust their labor supply well before reaching the SRA. Higher full-time participation around age 60 may therefore reflect precautionary behavior: workers increase earnings earlier to accumulate liquid savings, which allows them to bridge the cash-flow deficit before pension eligibility and smooth consumption over the extended pre-pension horizon.

Relative to cohort 2, cohort 6 is also more likely to work part-time from age 60 onward. Again, this pattern suggests an anticipatory adjustment in labor supply. Facing a higher SRA, individuals expect a longer working life and reduce hours earlier to sustain employment over this extended horizon. Due to the SRA reform, part-time employment becomes a preferred alternative to full retirement, allowing individuals to maintain a liquid income stream to smooth consumption in some form of phased retirement.

Overall, the patterns in Figure 1 suggest that cohorts adjust their late-career labor supply in different ways, reflecting whether institutional rules grant early access to occupational pension cash flow or delay public pension access via a higher SRA. For cohort 1, the adjustment shifts from working full-time to part-time: access to early occupational pensions relaxes the immediate cash-flow constraint, allowing them to smooth consumption when working part-time. For cohort 6, the adjustment instead shifts from retirement to working part-time: when full-time work is not feasible, part-time hours act as a liquidity bridge to smooth consumption until reaching the higher SRA.

More broadly, these abrupt behavioral responses suggest that a liquidity mechanism, rather than a lifetime wealth effect or a price incentive, drives the observed labor supply responses. If losing the early-retirement scheme acted primarily as a negative wealth shock, standard life-cycle theory predicts a gradual labor supply adjustment spread across remaining working years, not the sharp, sudden increase observed in cohorts 2-6. Similarly, because the post-reform system is actuarially neutral, it creates no discrete, localized jump in the relative price of leisure on an individual's 60th birthday. Instead, the gap between cohorts opens abruptly at age 56 — when cohort 1 accesses early-retirement benefits while cohorts 2-6 face an institutional lock on pension cash — and persists until age 60. Furthermore, cohorts 1 and 2 exhibit a sharp decline in full-time employment precisely on their 60th birthday. Because neither permanent lifetime wealth nor marginal price incentives change discretely on that exact day, this sudden kink suggests that the binding constraint was an inability to bridge immediate cash flows.

Figure 2 distinguishes between individuals who claim and do not claim partial pensions while working part-time.[3] In the upper panel of the figure, among cohorts 2-6, part-time employment combined with pension claiming rises steadily from age 60 and peaks between ages 63 and 65. This pattern suggests that partial pensions facilitate gradual retirement by bridging the immediate income gap at ages when the disutility of work rises and the actuarial penalty for early pension claiming diminishes. Like the younger cohorts, cohort 1 also uses partial pensions to combine part-time earnings with pension income. However, they do so substantially more often, particularly from age 60, when all cohorts become eligible for occupational pensions and are directly comparable, due to their access to generous early-retirement benefits.

In the lower panel of Figure 2, where part-time employment is not combined with pension claiming, cohort 1 engages in part-time employment less frequently than cohorts 2-6. This contrasts with the upper panel, where cohort 1 is more likely to work part-time while claiming a pension. This pattern suggests that part-time work is chosen by cohort 1 primarily when it is paired with pension liquidity, reinforcing our claim that partial pensions are used to relax a cash-flow constraint. To explore whether these patterns reflect a consumption-smoothing mechanism in partial retirement, we analyze further descriptive statistics.

[3] Under ABP pension regulations, once a worker reduces hours and chooses to claim a partial pension, the pension claim is irrevocable at subsequent ages. Accordingly, individuals shown in the upper panel, those who have claimed a partial pension, cannot reappear in the lower panel. In contrast, individuals who reduce their work hours without claiming a partial pension retain the option to do so later. Therefore, individuals in the lower panel may subsequently transition to the upper panel. Among those who initially forgo claiming (lower panel), only 18% eventually claim a partial pension at a later age (upper panel).

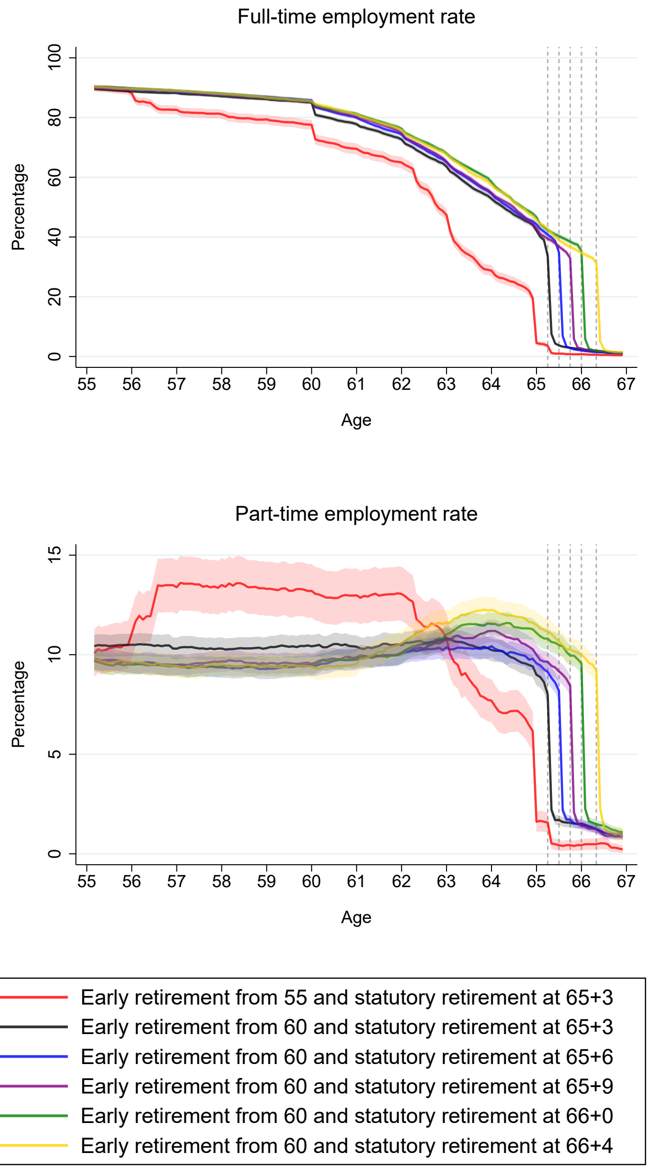



Figure 1: Fractions of individuals working full-time and part-time from age 55 to 68.

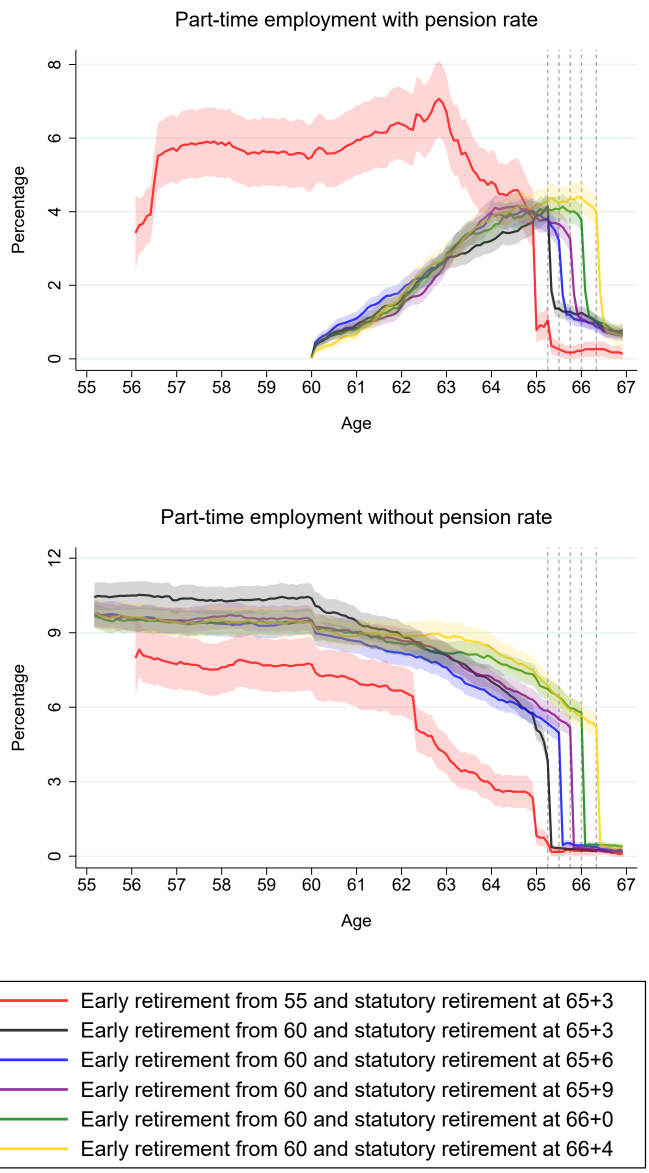



Figure 2: Fractions of individuals working part-time and claiming partial pensions (upper panel), and individuals working part-time without any claim of partial pensions (lower panel) from age 55 to 68.

We define partial retirement as the transition from full-time employment at age 55 years and 2 months to uninterrupted part-time work of any duration prior to full retirement. By only studying the part-time employment rates in Figure 1, we cannot distinguish between transitions into partial retirement and stable patterns of part-time work or temporary reductions in hours. In our sample, between age 55 years and 2 months and age 66 years and 11 months, 23% of individuals work part-time at some point. This 23% breaks down into 12% who meet our criteria for partial retirement, 8% who always work part-time, and 3% who alternate between part-time and full-time employment.

Table 2 shows descriptive statistics on partial retirement across birth cohorts. Approximately 12% of individuals partially retire from their full-time position at some point between age 55 years and 2 months and age 66 years and 11 months. This implies that much of the part-time employment observed in Figure 1 reflects participation in a partial-retirement trajectory. The oldest cohort exhibits the highest participation rate, enters partial retirement earlier, and remains in this state for more years, patterns consistent with their ability to afford partial retirement due to the generous occupational pension provisions available to them. Their substantially higher pension-claiming rate supports this interpretation, while their lower DI-claiming rate suggests that pension benefits may substitute for DI.

Table 2: Partial retirement statistics by birth cohort

| | Birth cohort | | | | | |
|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 |
| In partial retirement (%) | 0.13 | 0.11 | 0.11 | 0.12 | 0.12 | 0.12 |
| | (0.34) | (0.31) | (0.32) | (0.32) | (0.33) | (0.32) |
| Age at onset of partial retirement | 59.93 | 62.15 | 62.34 | 62.55 | 62.62 | 62.60 |
| | (3.38) | (2.76) | (2.67) | (2.58) | (2.66) | (2.66) |
| Age at end of partial retirement | 63.45 | 64.55 | 64.68 | 64.87 | 65.04 | 65.18 |
| | (1.93) | (1.80) | (1.79) | (1.72) | (1.83) | (1.85) |
| Duration of partial retirement (Yrs) | 3.52 | 2.40 | 2.35 | 2.32 | 2.42 | 2.59 |
| | (2.81) | (2.08) | (1.95) | (1.99) | (2.01) | (2.10) |
| Claiming pension during partial retirement (%) | 0.82 | 0.64 | 0.62 | 0.60 | 0.55 | 0.50 |
| | (0.38) | (0.48) | (0.48) | (0.49) | (0.50) | (0.50) |
| Claiming DI during partial retirement (%) | 0.07 | 0.12 | 0.09 | 0.09 | 0.09 | 0.09 |
| | (0.25) | (0.33) | (0.29) | (0.28) | (0.29) | (0.29) |

Note: Standard deviation in parentheses.

In Figure 3, we examine how individuals navigate the financial transition into partial retirement. We exploit differences in whether individuals claim pension benefits, DI benefits, or neither to identify the mechanisms that make partial retirement financially feasible. The top panel shows an obvious reduction in working hours, transitioning from full-time employment at the onset of partial retirement. Notably, however, the largest decline is observed among pension and DI recipients, suggesting that access to these benefits plays an important role in determining labor supply decisions. These groups reduce their hours more dramatically, as the supplementary income from benefits helps offset the loss of earnings. In contrast, those without benefit claims maintain higher number of work hours, relying more on labor income to smooth consumption.

The middle panel confirms the pattern observed in the top panel: while gross income declines across all groups, the reduction is less pronounced for pension and DI recipients. This suggests that benefit income helps offset the loss of labor earnings. These groups can therefore afford to work fewer hours during partial retirement, as their benefit income supplements part-time wages and supports continued consumption smoothing.

The bottom panel shows personal assets across the three groups. Individuals who do not claim benefits possess significantly more assets than those who do. This disparity may help explain their lower reliance on benefits: people with substantial savings or assets are less likely to seek income from benefits. Beyond these group-level differences, none of the three groups appear to decumulate assets during partial retirement to smooth consumption. This aligns with evidence from the Netherlands, where retirees often treat accumulated assets as a long-term reserve rather than a source of regular spending (Van Ooijen et al., 2015). This may help explain why wealthier individuals work more during partial retirement: since they neither rely on benefits nor draw down savings, they depend more heavily on labor income to finance consumption.

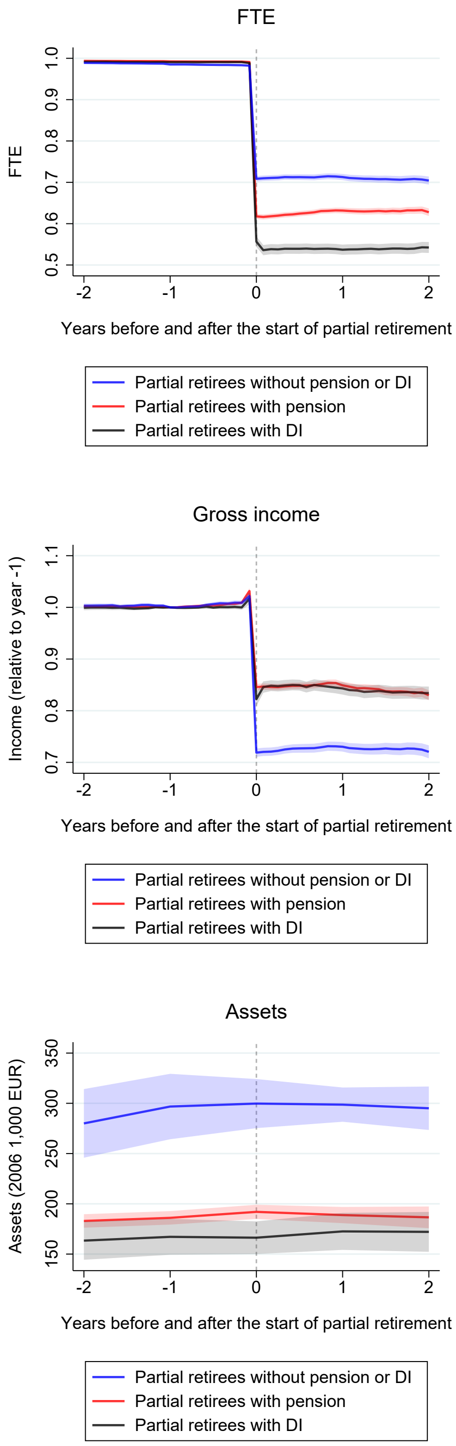



Figure 3: Labor Supply, earnings, and assets in the years before and after the start of partial retirement, by pension and DI claiming status.

# 5 Reduced-form analysis and results

This section documents the reduced-form effects of the two pension reforms. The aim is twofold. First, we use a DiD design to provide transparent, model-free evidence on how individuals adjusted their employment trajectories when early-retirement pathways were abolished and when the SRA was raised. In particular, we test whether behavioral adjustments occur discretely at pension eligibility thresholds, which would indicate that liquidity constraints play a more important role in shaping late-career labor supply relative to standard wealth and price effects. Second, these reduced-form patterns serve as an empirical benchmark for the structural analysis that follows. In particular, the behavioral responses uncovered here motivate the key mechanisms embedded in the dynamic life-cycle model and inform the choice of targeted moments used in the Method of Simulated Moments (MSM) estimation.

## 5.1 DiD model

Building on the descriptive patterns illustrated in Figure 1, we formalize labor supply responses to the two major pension reforms taking a DiD approach.[4] Specifically, we estimate the following linear probability model:

$$y_{it} = \alpha_i + \sum_{k=1}^{24} \beta_k \left( Treated_i \times d_k(t) \right) + \sum_{k=1}^{24} \gamma_k d_k(t) + \varepsilon_{it}. \tag{1}$$

$i$ indexes individuals. $t$ indexes age in months, beginning at 55 years and 2 months and extending to 66 years and 11 months. $k$ denotes age categories defined in six-month intervals. Because the data begin at 55 years and 2 months (Table 1), the first category is shorter and covers ages 55 years and 2 months to 55 years and 5 months. The final category covers ages from 66 years and 6 months to 66 years and 11 months. We conduct the analysis by age in six-month intervals as finer distinctions are not meaningful for our purposes.

$\alpha_i$ captures individual fixed effects. $Treated_i$ is a binary indicator equal to one for individuals in the treatment group. The dummy $d_k(t)$ identifies age category $k$. The coefficients $\beta_k$ measure the treatment effect at each age category relative to the omitted baseline category, defined as the age group immediately preceding the reform announcements. The outcome variable $y_{it}$ is a binary indicator for either full-time or part-time employment. Standard errors are clustered at the individual level.

## 5.2 Reduced-form estimates

In the left panel of Figure 4, we examine the impact of abolishing the early retirement scheme. We consider two adjacent birth cohorts: individuals born between November 1 and December 31, 1949, and those born between January 1 and September 30, 1950 (birth cohorts 1 and 2 in Table 1). While both cohorts face the same SRA, they differ in their access to generous early retirement pensions, allowing us to estimate the effect of changing early retirement incentives on labor market participation. To isolate the causal effect of the reform, we restrict the control group to individuals born in December 1949, the final month of the earlier cohort, and define the treatment group as those born in January 1950, the first month of the later cohort. This narrow comparison minimizes cohort-related confounding and ensures that both groups are similar in observable characteristics except for their exposure to the reform.

[4] In Appendix B, we present results from a regression discontinuity (RD) approach and show that both identification strategies yield the same qualitative conclusions for all outcomes, as well as similar magnitudes of the reform effects.

We use the reform announcement date, July 7, 2005, to define the pre- and post-reform periods. On this date, the control group was 55 years and 7 months old, while the treatment group was 55 years and 6 months old. Accordingly, we define the pre-treatment period as being younger than 55 years and 6 months, and the post-treatment period as being at this age or older. Before turning 55, neither group is eligible to claim occupational pension rights (Section 2); after this age, only the control group is eligible for generous early retirement provisions from their occupational pension fund.

As already shown in Figure 1, the two groups exhibit nearly identical full- and part-time employment rates during age 55, a period during which neither cohort is eligible for generous occupational pensions. This results in a zero treatment effect prior to eligibility and lends support to the common trends assumption. Beginning at age 56, when the control group becomes eligible for generous occupational pensions, the full-time employment rate starts to diverge. This divergence becomes especially pronounced from age 62 onward. When both groups reach the SRA, their full-time employment rates converge leading to no treatment effect. As discussed in relation to Figure 1, these patterns are consistent with institutional pension eligibility thresholds and actuarial rules, and highlight the strong influence of financial incentives on continued labor market engagement.

The estimates for part-time employment also mirror the patterns shown in Figure 1. At earlier ages, the treatment effects are negative, indicating that generous pension provisions allow individuals in the control group to afford partial retirement. After age 62, when claiming pensions no longer triggers actuarial penalties, full retirement becomes financially attractive, leading to a shift away from part-time work. The wide confidence intervals reflect the small sample size, as the analysis compares cohorts born within a month across control and treatment groups.

In the right panel of Figure 4, we examine the impact of increasing the SRA. We consider two birth cohorts: individuals born between January 1 and September 30, 1950, and those born between January 1 and July 31, 1953 (birth cohorts 2 and 6 in Table 1). These cohorts do not have access to generous early-retirement provisions, while cohort 6 reaches eligibility for the state pension and actuarially neutral occupational pensions at an age that is 1 year and 1 month later. To isolate the causal effect of the reform, we restrict the control group to individuals born in January 1950, the first month of the earlier cohort, and define the treatment group as those born in January 1953, the first month of the later cohort. We define the treatment group as individuals born in January 1953 rather than July 1953, because the former cohort can be followed up to age 66 years and 11 months, whereas the latter is observed only up to age 66 years and 5 months (Table 1). By selecting cohorts furthest apart, we maximize the age contrast at pension eligibility thresholds.

We use the reform announcement date, June 10, 2011, to define the pre- and post-reform periods. On this date, the control group was 61 years and 5 months old, while the treatment group was 58 years and 5 months old. Accordingly, we define the pre-treatment period as being younger than 58 years, and the post-treatment period as being at this age or older. Prior to reaching age 58, neither group is aware that their SRA will be increased; after this point, anticipatory responses become more likely for the treatment group, whose SRA is higher.

Again, the estimates align with the patterns shown in Figure 1. The treatment effect is statistically indistinguishable from zero prior to the treatment cut-off, supporting the common trends assumption. By age 60, the treatment group already exhibits higher rates of full-time employment than the control group. The rise in full-time work likely reflects precautionary behavior, with individuals working more to build financial security in anticipation of early exit before reaching their higher SRA. In contrast, the treatment effect on part-time work at later ages indicates that younger cohorts respond by shifting toward partial retirement to accommodate a longer expected working life.

Taken together, these reduced-form estimates demonstrate that labor-supply adjustments occur discretely at exact pension eligibility thresholds rather than smoothly over time. This localized, discontinuous response confirms that labor supply reacts directly to immediate liquidity availability rather than following smooth, unconstrained life-cycle adjustments, providing the empirical pattern that motivates our structural framework.

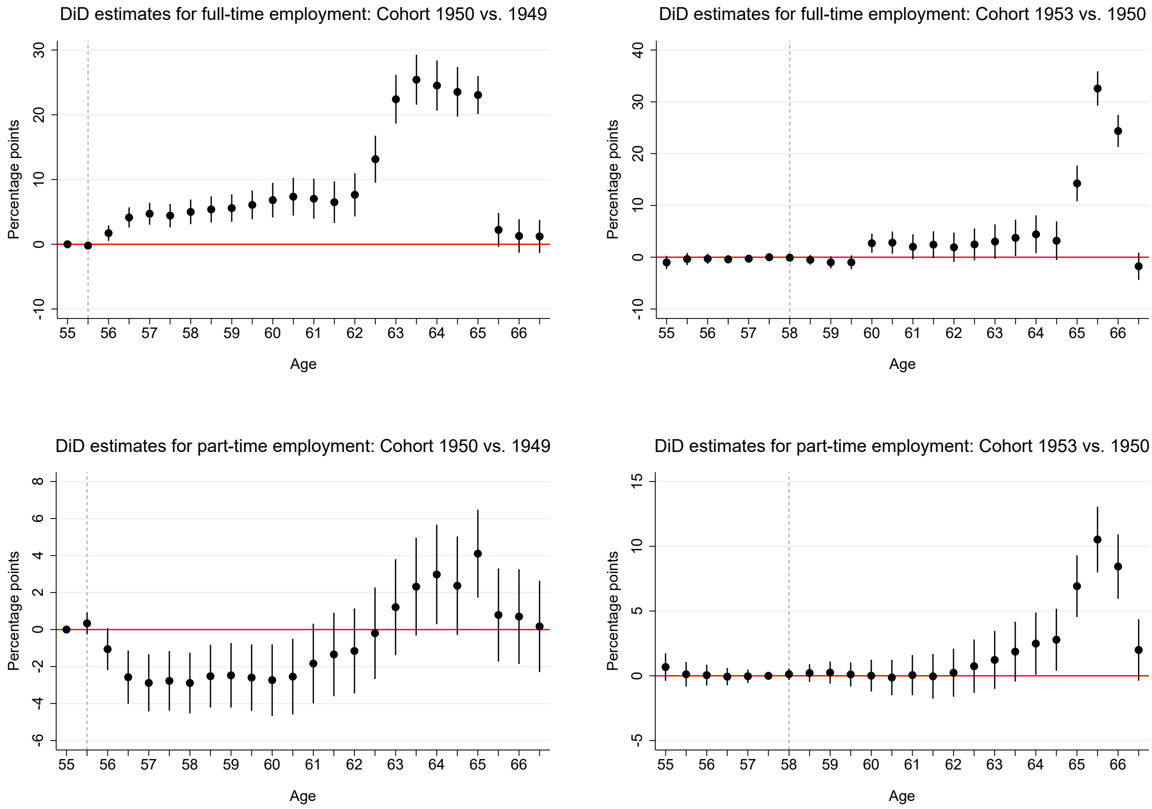



Figure 4: Impact of the early retirement (left panel) and SRA (right panel) reforms.

# 6 Dynamic model of labor supply, pension claiming, and consumption

In our exploratory and reduced-form analyses in Sections 4 and 5, we relied on both exogenous changes in pension incentives and individuals' endogenous choices to study partial pension claiming behavior. Taken together, these analyses provided suggestive evidence that liquidity constraints limit partial retirement. However, reduced-form analysis cannot isolate liquidity constraints from the wealth and price effects that could occur simultaneously during pension reforms. To disentangle these competing mechanisms, we develop a dynamic programming life-cycle model of labor supply, (partial) pension claiming, and savings. Our model controls for standard financial incentives within the agent's dynamic budget set, allowing liquidity frictions to be identified. First, price effects are accounted for by embedding the full non-linear tax and social insurance system alongside actuarial pension adjustment rules directly into the budget constraint, pinning down the exact marginal financial return to working full-time versus part-time at each age. Second, lifetime wealth effects are accounted for by tracking asset accumulation and projected lifetime pension entitlements over time. Having accounted for these price and wealth channels, the model isolates liquidity frictions through the non-negativity constraint on assets, which restricts consumption smoothing prior to pension eligibility independently of standard financial incentives. Ultimately, recovering the structural parameters of this model allows us to perform counterfactual policy simulations that demonstrate how access to pension claiming relaxes liquidity constraints to shape partial retirement decisions.

The remainder of this section lays out the structural model. We begin by describing the agent's decision problem, specifying how forward-looking individuals trade off consumption and leisure across periods subject to endogenous and exogenous state variables. Next, we present our estimation strategy and detail how the preference parameters are identified and recovered from administrative data. Our specification builds on leading life-cycle models of retirement, including De Bresser (2024), De Nardi et al. (2025), and French et al. (2022). For ease of reference, Tables 5 and 6 in Appendix C.1 provide a complete summary of the notation.

## 6.1 Agent's optimization problem

In this section, we first define the agent's decision and state variables, and then specify the functional forms for preferences, budget constraints, and the stochastic processes that govern the environment.

### 6.1.1 Decision and state variables

We begin by defining the agent's decision variables, state variables, and payoff variable, which together determine the feasible choices, the evolution of the state, and the flow return in each period.

#### Decision variables

Agents are indexed by $i$. Time is indexed by $t$, which is discrete and represents age in years. The description of the environment in period $t$ for agent $i$ consists of three components. The first component is a vector of decision variables, $d_{it} = \{h_{it}, o_{it}, c_{it}\}$. The earliest age at which an agent makes decisions is $t = t_0$, and the maximum possible age is $t = 100$. We set $t_0 = 56$, since this is the first period in which asset data are available for birth cohorts 1 and 2 (Section 3). Each agent faces an exogenous probability of death that increases with age and reaches one at $t = 100$.

The variable $h_{it}$ represents annual hours of work and takes values from the baseline choice set $\mathcal{H} = \{0, 1000, 1600, 2000\}$. A value of 0 denotes non-employment. We treat non-employment as a voluntary choice, implying that involuntary job separation is not allowed in the model. Accordingly, we exclude unemployment, which we assume to be involuntary. This assumption is not restrictive, as at most 1% of individuals in our sample receive unemployment insurance at any age. A value of 2000 represents full-time work, following the literature (French, 2005; French and Jones, 2011). The intermediate values of 1000 and 1600 hours correspond to 0.5 and 0.8 FTE, respectively, the two most common part-time arrangements observed in the data (Figure 13). We assume that there are no demand-side constraints on reducing work hours. This assumption is consistent with the Dutch Flexible Working Act, which grants employees the right to request a reduction in working hours (Section 2).

On the other hand, the baseline choice set $\mathcal{H}$ is restricted in two respects and implies that the feasible choices at a given age constrain future choices. First, individuals cannot work after reaching the age of state pension eligibility, as employment contracts are terminated at that point in the Netherlands. Second, hours of work cannot increase with age, which means that the choice set is restricted by past decisions. These assumptions imply that retirement is an absorbing state: once retired, individuals cannot transition back into employment. Since upward shifts in hours are rare in the data, we impose this monotonicity restriction to reduce the computational burden.

The variable $o_{it} \in \mathcal{O} = \{0, 1\}$ denotes the binary decision to claim occupational pension rights. The minimum claiming age for these rights is cohort-specific (Table 1). We assume that all individuals claim their accrued occupational pension rights no later than the state pension age. This assumption is supported by our observation that in the data nearly all individuals claim occupational pension rights once they become eligible to claim accrued state pension rights (Figure 15).

The claiming decision for occupational pension rights is independent of the claiming of state pension rights, which are automatically paid at the SRA. Under ABP regulations, the occupational-pension claiming decision is irreversible. Consequently, as with the hours choice set $\mathcal{H}$, current choices from the feasible set $\mathcal{O}$ affect the future choice set $\mathcal{O}$.

Under ABP regulations, individuals may claim a fraction of accrued occupational pension rights while working, up to a maximum of $(1 - \text{FTE})$. We assume that, conditional on the pension claiming decision and a given labour supply choice, individuals claim the maximum rate permitted by regulation. This assumption is strongly supported by observed behavior in the data (Figure 14). Accordingly, we allow agents to choose between two partial-retirement arrangements: working 0.5 FTE while claiming 50% of pension rights, or working 0.8 FTE while claiming 20%.

Agents weigh labor income against pension entitlements across time and allocate resources between consumption and savings. The last decision variable, consumption, is denoted by $c_{it} \in \mathcal{C}(h_{it}, o_{it}, x_{it})$. We assume a continuous consumption choice set, which allows any positive level of consumption but is constrained by available resources. In particular, the maximum feasible level of consumption depends on income and savings, as determined by current choices $h_{it}$ and $o_{it}$ together with the state variables introduced below. Borrowing is not permitted.

#### State variables

The second component of the description of the environment in period $t$ for agent $i$ is the vector of state variables, $x_{it} = \{t, w_{it}, p_{it}, s_{it}, k_i, a_{it}, m_{it}\}$. The variable $w_{it}$ is continuous and represents the annual gross full-time wage available to the agent in period $t$ if the agent chooses to work. It is treated as an exogenous variable, since wages are determined outside the agent's decision process.

$p_{it}$ is continuous and represents accrued pension rights at the start of period $t$. It is endogenous because it depends on the agent's labour-supply choices and the pension premiums paid using the pension contribution base.

$s_{it}$ represents retirement status. It is an endogenous state variable, determined by the agent's past labour supply and pension claiming decisions, $h_{it-1}$ and $o_{it-1}$. Since both decisions are discrete, $s_{it}$ is discrete as well.

$k_i$ indicates the agent's birth cohort, determined by date of birth. Since pension rules vary across cohorts, $k_i$ serves as the exogenous variable that governs which pension regime applies to the agent.

$a_{it}$ is continuous and represents the assets available at the start of period $t$. It accumulates endogenously, as it depends on, among others, past savings and consumption decisions.

$m_{it}$ is a binary indicator of health status, distinguishing between good and bad health. It is treated as an exogenous variable, since health is taken as given by the agent.

$t$, $s_{it}$ and $k_i$ together constrain the baseline choice sets $\mathcal{H}$ and $\mathcal{O}$ so that $h_{it} \in \mathcal{H}(x_{it})$ with $\mathcal{H}(x_{it}) \subseteq \mathcal{H}$ and $o_{it} \in \mathcal{O}(x_{it})$ with $\mathcal{O}(x_{it}) \subseteq \mathcal{O}$. The entire vector of state variables $x_{it}$ determines the upper bound of the consumption choice set $\mathcal{C}$, such that the agent's consumption decision satisfies $c_{it} \in \mathcal{C}(h_{it}, o_{it}, x_{it})$.

#### Payoff variable

The third component, $\tau(d_{it}, x_{it})$, is the payoff variable. It maps the decision and state variables into current-period net income by aggregating earnings, disability benefits, and pension benefits, and subtracting income taxes, asset taxes, and social security contributions. This net-income payoff constitutes the flow return in period $t$, determining both the current-period reward that enters the Bellman equation and the immediate consumption-leisure trade-off faced by the individual.

### 6.1.2 Preferences and functional forms

Having introduced the agent's decision and state variables, we now specify the preferences and functional forms that govern utility, income processes, and constraints.

#### Preferences

Agents derive utility from consumption and leisure according to

$$u(c_{it}, l_{it}) = \frac{1}{\lambda} c_{it}^{\lambda} + \psi \frac{1}{\gamma} l_{it}^{\gamma}, \tag{2}$$

where $c_{it}$ denotes annual consumption and $l_{it}$ denotes leisure. The parameters $\lambda$ and $\gamma$ govern the curvature of utility with respect to consumption and leisure, respectively. $\psi$ scales the weight of leisure in utility relative to consumption and therefore shifts the marginal rate of substitution between leisure and consumption.[5]

Following French (2005) and French and Jones (2011), leisure is defined by the time-budget constraint, which states that individuals have a fixed yearly endowment of 4,000 hours that

[5]This functional form of utility is similar to Gustman and Steinmeier (2005) and Keane and Wasi (2016). Assuming non-separable utility in consumption and leisure results in a worse model fit to the data, especially with respect to assets accumulation over time.

must be allocated between work and leisure. Bad health reduces this endowment by $\delta$ hours, so leisure is given by

$$l_{it} = \frac{4,000 - h_{it} - \delta\, m_{it}}{4,000}. \tag{3}$$

Here, $h_{it}$ denotes annual hours worked, and $m_{it}$ is a binary indicator of bad health as defined in the preceding section. $\delta$ is the health time penalty parameter, representing the reduction in the yearly time endowment available for leisure when an individual is in bad health. We expect the estimate of parameter $\delta$ to be positive, indicating that bad health reduces the effective time endowment available for leisure and thereby increases the marginal utility of leisure relative to healthy individuals. This formulation implies that leisure $l_{it}$ is a continuous variable bounded between 0 and 1, representing the share of the yearly time endowment allocated to leisure.

Following De Nardi (2004), agents also derive utility from leaving a bequest, modeled as

$$q(a_{i,t+1}) = \frac{\rho_1}{\lambda}\left(\rho_2 + a_{i,t+1}\right)^{\lambda}, \tag{4}$$

where $a_{i,t+1}$ denotes the assets passed on if death occurs at the end of period $t$. The parameter $\rho_1$ governs the overall strength of the bequest motive relative to utility from consumption and leisure, while $\rho_2$ shifts the function and influences its curvature.

### Wages

Among the state variables in the vector $x_{it}$, wage $w_{it}$ and health status $m_{it}$ are the only two that evolve stochastically. We assume that the logarithm of gross full-time wage, $\ln(w_{it})$, follows an autoregressive process:

$$\ln(w_{it}) = \alpha + \phi \ln(w_{i,t-1}) + \epsilon_{it}, \tag{5}$$

where current wage depends on last period's wage and an idiosyncratic shock. The error term is assumed to be normally distributed, $\epsilon_{it} \sim N(0, \sigma_\epsilon^2)$, and independently and identically distributed across individuals and time.

These modeling assumptions are consistent with the institutional context faced by Dutch public sector employees. Wages are determined by collective labour agreements, which establish salary scales prescribing increases with years of tenure and leaving little scope for individual bargaining. In practice, most individuals in our sample have likely reached the top of their wage scale by the age of 56, when they enter the model. Initial wages $w_{i,56}$ can therefore capture heterogeneity across dimensions such as education or occupation, but their subsequent law of motion is assumed to depend only on last-period wage and not on other state variables.

Full-time wages $w_{it}$ enter the agents' maximization problem through earnings $y_{it}$, which are proportional to hours worked:

$$y_{it} = f_{it} \cdot w_{it} \cdot \exp[\log(\eta)\, I\{f_{it} < 1\}], \tag{6}$$

where $f_{it}$ denotes FTE, defined as $f_{it} = \frac{h_{it}}{2{,}000}$. Earnings incorporate a penalty for part-time work, consistent with evidence from the literature (see, e.g., Russo and Hassink, 2008 for the Netherlands). In this formulation, the parameter $\eta \in (0, 1)$ determines the magnitude of the penalty, with $1 - \eta$ representing the proportional earnings loss associated with part-time employment.

**Health**

The second stochastic state variable, $m_{it}$, is a binary indicator of bad health. Its dynamics are modeled as a logistic probability that depends on age and last period's health status:

$$\Pr(m_{it} = 1 \mid m_{i,t-1}, t) = \frac{\exp(\mu_0 + \mu_1 t + \mu_2 m_{i,t-1})}{1 + \exp(\mu_0 + \mu_1 t + \mu_2 m_{i,t-1})}. \tag{7}$$

The parameters $\mu_0$, $\mu_1$, and $\mu_2$ govern the baseline probability, the effect of age, and the persistence of health status, respectively. This specification allows the likelihood of poor health to vary with age and to depend on whether the individual was already unhealthy in the previous period.

The health status affects agents' choices in three ways. First, health directly influences the utility derived from leisure, as specified in equation(3).

Second, individuals in bad health receive disability insurance (DI) benefits, so health status affects income directly. In the Netherlands, DI pays 70% of pre-sickness wages multiplied by a disability grade, where the disability grade equals one minus the remaining work capacity expressed in FTE. Eligibility requires a disability grade of at least 35%. In the sample data, 75% of DI recipients have a disability grade below 80%, meaning they are partially disabled, and 50% of recipients fall between 35% and 65%. For tractability, we assume that all individuals in bad health ($m_{it} = 1$) have a disability grade of 50%. DI benefits $b_{it}$ are therefore modeled as

$$b_{it} = m_{it} \cdot 0.7 \cdot w_{it} \cdot 0.5. \tag{8}$$

Third, health restricts the feasible choice set for hours worked. Specifically, $\mathcal{H}(X_{it}) = \{0, 1000\}$ if $m_{it} = 1$, meaning that sick individuals can supply at most 1,000 hours per year (FTE = 0.5). This assumption is consistent with assigning all individuals in bad health a disability grade of 50%.

**Retirement status**

Retirement status $s_{it}$ is a discrete endogenous state variable that evolves deterministically based on the individual's labour supply and pension-claiming decisions in the previous period, $h_{i,t-1}$ and $o_{i,t-1}$. Table 3 illustrates how each combination of hours worked and pension claiming decision translates into one of eight possible retirement states in the following period. Note that state $s_{it} = 2$ cannot occur, since individuals are not permitted to work full-time while simultaneously claiming pension.

This implies that $h_{it}$ and $o_{it}$ jointly determine $s_{i,t+1}$, which in turn defines the feasible choice sets of hours of work and pension claiming in the next period, $\mathcal{H}(x_{i,t+1})$ and $\mathcal{O}(x_{i,t+1})$. As described in the preceding section, we assume that hours worked cannot increase with age. Consequently, the feasible choice set $\mathcal{H}(x_{it})$ depends on $s_{it}$ as follows:

$$\mathcal{H}(x_{it}) = \begin{cases} \{0, 1000, 1600, 2000\}, & s_{it} = 1, \\ \{0, 1000, 1600\}, & 3 \leq s_{it} \leq 4, \\ \{0, 1000\}, & 5 \leq s_{it} \leq 6, \\ \{0\}, & 7 \leq s_{it} \leq 8. \end{cases}$$

Similarly, pension claiming is an absorbing decision: once agents begin claiming, they must continue to do so. This implies the following feasible claiming sets:

$$\mathcal{O}(x_{it}) = \begin{cases} \{0, 1\}, & s_{it} \text{ is odd}, \\ \{1\}, & s_{it} \text{ is even}. \end{cases}$$

Table 3: Mapping of labour supply and pension claiming at $t$ to retirement status in $t+1$

| $h_{it}$ | $o_{it}$ | $s_{i,t+1}$ |
|---|---|---|
| 2,000 | 0 | 1 |
| 2,000 | 1 | 2 |
| 1,600 | 0 | 3 |
| 1,600 | 1 | 4 |
| 1,000 | 0 | 5 |
| 1,000 | 1 | 6 |
| 0 | 0 | 7 |
| 0 | 1 | 8 |

### Occupational pension rights

Occupational pension rights $p_{it}$ are modeled as an endogenous state variable that evolves according to the accrual rules of the ABP scheme as described in Kantarcı et al. (2013):

$$p_{i,t+1} = p_{it} + f_{it} \cdot z(k_i) \cdot \big(w_{it} - g(k_i)\big). \tag{9}$$

Pension rights accrue over pensionable income, defined as the full-time wage $w_{it}$ net of the state pension offset $g(k_i)$. This offset reflects that participants simultaneously build entitlements under the public old-age pension (AOW). Because contributions to the occupational plan are proportional to full-time equivalence, pension rights accumulate in proportion to $f_{it}$. Accrual occurs at rate $z(k_i)$, which is calibrated by the pension fund to deliver a targeted replacement rate at retirement. Both $g(k_i)$ and $z(k_i)$ vary across birth cohorts due to the early retirement reform implemented in 2006:

$$g(k_i) = \begin{cases} 15,500, & k_i \leq 1949, \\ 9,600, & k_i \geq 1950, \end{cases}$$

and

$$z(k_i) = \begin{cases} 1.75\%, & k_i \leq 1949, \\ 2.05\%, & k_i \geq 1950. \end{cases}$$

Before retirement, $p_{it}$ represents only accumulated rights, a proposal of how much could eventually be received. It is only at the point of claiming that these rights are transformed into the actual annuity entitlement $e_{it}$, which determines the income stream in retirement:

$$e_{it} = o_{it} \cdot (1 - f_{it}) \cdot p_{it} \cdot v(d_{it}, x_{it}). \tag{10}$$

This entitlement reflects the individual's claiming choice, captured by the binary indicator $o_{it}$, and is adjusted for partial pension claiming through the full-time equivalent factor $(1-f_{it})$. The actuarial adjustment function $v(d_{it}, x_{it})$ adjusts the payout to account for the age at claiming. By contrast, state pension benefits $g_{it}$ are automatically paid at the cohort-specific SRA.

### Budget constraint

Next-period assets are determined by the budget constraint

$$a_{i,t+1} = \big(a_{it} + \tau(d_{it}, x_{it}) - c_{it}\big)(1 + r), \tag{11}$$

where $\tau(d_{it}, x_{it})$ denotes net income. Net income incorporates earnings, disability benefits, occupational and state pension entitlements, social security and national insurance contributions, pension contributions, income and asset taxes, and applicable tax credits. The inner term $a_{it} + \tau(d_{it}, x_{it}) - c_{it}$ represents end-of-period assets before returns, and multiplying by $(1 + r)$ yields next-period assets under a constant net return $r$ on the asset stock.

In contrast to most dynamic programming models of retirement, our framework embeds the full set of institutional rules governing social security and national insurance contributions, taxation, and pension accrual. This allows the model to capture the precise incentives individuals face in the Dutch system. All institutional details are documented in Appendix C.2.

**Assets and borrowing condition**

Assets $a_{it}$ are an endogenous state variable evolving deterministically according to (11). We impose a no-borrowing condition,

$$a_{i,t+1} \geq 0, \tag{12}$$

for all $t$, so that assets cannot fall below zero. The budget constraint together with the no-borrowing condition (12) determines the feasible consumption set $\mathcal{C}(x_{it})$. The feasible asset set is $\mathcal{A} = \mathbb{R}_+$, so that $a_{it} \in \mathcal{A}$ for all $t$.

**Feasible decision set**

Given the state vector $x_{it}$ and the restrictions on labour supply, pension claiming, consumption, and borrowing, the feasible decision set in period $t$ is

$$\mathcal{D}\left(x_{it}\right) = \left\{\left(h_{it}, o_{it}, c_{it}\right) \mid h_{it} \in \mathcal{H}\left(x_{it}\right),\ o_{it} \in \mathcal{O}\left(x_{it}\right),\ c_{it} \in \mathcal{C}\left(h_{it}, o_{it}, x_{it}\right)\right\}.$$

### 6.1.3 Dynamic optimization problem

The agent's dynamic optimization problem can be expressed in recursive form as:

$$V_{it}\left(x_{it}\right) = \max_{d_{it} \in \mathcal{D}(x_{it})} \left\{u\left(c_{it}, l_{it}\right) + \pi_t \beta\, \mathbb{E}\left[V_{i,t+1}(x_{i,t+1}) \,\middle|\, x_{it}, d_{it}\right] + (1 - \pi_t) q(a_{i,t+1})\right\}. \tag{13}$$

Here, $\pi_t$ denotes the probability of surviving period $t$ conditional on survival in period $t - 1$. This survival probability is exogenous, decreases with age, and is constant across individuals, with $\pi_{100} = 0$. The parameter $\beta \in [0, 1]$ captures time preferences. Expectations $\mathbb{E}[\cdot]$ are taken with respect to future wage and health shocks, conditional on current states and choices. Since the problem does not admit a closed-form solution, we rely on numerical methods to obtain the value function.

## 6.2 Econometrician's estimation problem

We estimate the model in three steps. In the first two steps we recover parameters that can be identified without solving the full dynamic programming problem, either because they are well established in the literature or because they can be estimated directly from the data under our maintained assumptions. In the third step, we estimate the structural preference parameters that cannot be identified outside the model and therefore require solving the full dynamic programming problem.

In the first step, we set several parameters to values reported in the literature. Specifically, we fix the time discount factor $\beta$ at 0.99, consistent with the parameter estimates in De Bresser

(2024). The interest rate $r$ is set to 1%, reflecting the low interest rate environment in the Netherlands between 2006 and 2019, the first and last years of our sample period. We set the part-time wage penalty $1 - \eta$ to 13%, in line with Russo and Hassink (2008) and Keane and Wasi (2016). Conditional survival probabilities $\pi_t$ are taken from the 2006 life tables published by the Dutch Royal Actuarial Society.

In the second step, we estimate the parameters governing the exogenous evolution of wages and health status, which we assume to be independent of individual choices. Using the laws of motion in equations (5) and (7), we estimate the wage process with a linear regression and the health process with a binary logit model. The estimated wage and health processes generate wage profiles and health trajectories when simulating life-cycle histories. Estimation details and results are presented in Appendices D.1 and D.2.

In the third step, we estimate the vector of preference parameters $\theta := \{\lambda, \gamma, \psi, \delta, \rho_1, \rho_2\}$ using the MSM. The MSM estimator selects the preference parameters that generate simulated life-cycle decision profiles that best match the empirical moments, as evaluated by a GMM criterion function. Because the optimal weighting matrix performs poorly in small samples, we follow Altonji and Segal (1996) and use a diagonal weighting matrix containing the inverse of the estimated variances of the sample moments. Additional details on the MSM estimator are provided in Appendix D.3.

For each age and birth cohort, the MSM estimator targets the following empirical moments: average assets; pension claiming rates; full-time and part-time employment rates among healthy individuals; employment rates among sick individuals; and the share of individuals claiming pension benefits while working part-time.[6] Appendix D.4 provides further details on the preparation of the asset data used to construct the asset-related moments.

In Section 5, we use a DiD strategy to estimate the reduced-form effects of pension reforms on retirement behavior. The reforms vary across birth cohorts: individuals born in December 1949 are the only cohort eligible for the generous early occupational pension provisions, with eligibility starting at age 55, while those born in January 1950 and January 1953 differ in their SRA. These cohort-specific policy changes provide the exogenous variation needed for the reduced-form analysis.

In the structural analysis, we exploit the same cohort-specific reforms in a different way. Our dynamic life-cycle model is estimated using the MSM, and identification of the preference parameters comes from the behavioral differences induced by the SRA reform. We therefore use data from the January 1950 and January 1953 cohorts, which faced different SRAs but were otherwise subject to the same institutional environment. We construct initial state variables $(x_{i,t_0})$ and compute empirical moments using only these two cohorts, and the MSM criterion matches these moments to their simulated counterparts.

We reserve the December 1949 cohort for model validation. Validation refers to an out-of-sample test of the model: after estimating the preference parameters using only the SRA reform, we keep these parameters fixed and simulate the model under the early retirement rules that apply to the December 1949 cohort. We then compare the model's predictions to the observed behavior of this cohort. Because the early retirement reform involved substantially larger changes in pension rules and generated stronger behavioral responses, it provides a more demanding test of whether the estimated model can replicate behavior under a policy change that was not used for estimation.

As a robustness check, we reverse the roles of the two reforms. We estimate the model using the early retirement reform, using December 1949 and January 1950 cohorts, and validate it using the SRA reform. The resulting parameter estimates, presented in Appendix D.5, are very similar. This separation between estimation and validation ensures that the preference

[6]The MSM estimator targets 115 moments: 55 moments for the 1950 cohort and 65 for the 1953 cohort.

parameters are identified by one policy change while the model's predictive performance is assessed using another.

# 7 Estimation results and model validation

This section presents the estimates of the structural parameters and evaluates how well the model fits the data.

## 7.1 Parameter estimates and model fit

Table 4 presents the preference parameter estimates. Although specific to our context and functional-form assumptions, these estimates are broadly consistent with earlier studies. We estimate $\lambda$, which determines the marginal utility of consumption, to be approximately $-0.364$. Given that the elasticity of intertemporal substitution for consumption is defined as $\frac{1}{1-\lambda}$, this estimate implies a consumption EIS of 0.73. This is close to recent empirical evidence for the Netherlands by Marenčák and Nghiem (2025), who report an elasticity of 0.70. An EIS of this magnitude indicates a moderate preference for consumption smoothing across periods; individuals favor a relatively stable standard of living and resist sharp, uncompensated fluctuations in their expenditures over time. Consequently, this preference structure implies that individuals will actively avoid large drops in consumption unless faced with substantial financial shifts.

We estimate $\gamma$, which determines the marginal utility of leisure, to be $-0.066$. Given that the elasticity of intertemporal substitution for leisure is defined as $\frac{1}{1-\gamma}$, this estimate implies a leisure EIS of 0.94. An EIS of this magnitude indicates a high willingness to substitute leisure across periods; individuals are flexible in when they consume leisure and do not require gradual reductions in hours. This preference structure is consistent with abrupt retirement transitions.

Our estimated preference for abrupt retirement is seemingly paradoxical when contrasted with the moderate preference for consumption smoothing. In a standard life-cycle model without liquidity constraints, the combination of concave utility — implying a preference for both consumption and leisure smoothing — and a convex budget set implies that the optimal labor supply path is smooth: individuals gradually reduce hours while using savings to maintain a stable consumption profile. We argue that abrupt retirement arises because liquidity constraints make gradual retirement infeasible. Retiring abruptly at the pension eligibility age allows individuals to replace full-time earnings with pension income that is close to their previous wage level, thereby maintaining a smooth consumption path. In contrast, retiring gradually entails earning part-time wages, which fall short of sustaining pre-retirement consumption levels if individuals lack alternative income sources to supplement part-time earnings.

Rogerson and Wallenius (2013) show that in a standard life-cycle model with flexible hours, abrupt retirement is optimal only if individuals are highly willing to substitute leisure over time. In their framework, explaining abrupt retirement requires a high leisure EIS. However, by introducing budget set nonconvexities, such as fixed costs of working or part-time wage penalties, abrupt retirement becomes easier to rationalize even with lower elasticities. Ameriks et al. (2020) provide empirical support for this prediction. Using stated-preference data, they find a modest late-life EIS, suggesting that demand-side constraints and production nonconvexities, rather than a high willingness to substitute, drive the observed prevalence of abrupt retirement.

Nonconvexities in the budget set are unlikely to be the driver of abrupt retirement in our context. Stated-preference evidence from Kantarcı et al. (2025) shows that more than one in three individuals in the Netherlands would prefer a gradual transition if such an option were unconstrained. Moreover, institutional features of the Dutch labor market suggest that demand-side constraints are unlikely to be binding. Employees hold a statutory right to request reduced working hours, which employers must honor unless they can demonstrate compelling business interests for refusal. This legal framework facilitates gradual retirement, suggesting that the observed abrupt transitions into retirement are driven by factors other than nonconvex budget sets or a restricted set of feasible hours.

In Section 8, we conduct policy simulations to show that when institutional barriers are relaxed, specifically by allowing part-time wages to be supplemented with alternative sources of income, partial retirement becomes attractive. These results confirm that the observed abrupt transitions are a response to liquidity constraints rather than an inherent distaste for partial retirement.

The parameter $\psi$, which scales the weight of leisure in utility relative to consumption, is estimated to be 0.041. The order of magnitude of our estimate also reflects the normalization adopted in the model: leisure $l_{it}$ is set to 0.5 for a healthy full-time worker, while simulated annual consumption $c_{it}$ is approximately 30,000, so the leisure term must be scaled accordingly to have a comparable impact on utility.

The time cost of bad health, $\delta$, is estimated to be about 587 hours per year. This is somewhat larger but still comparable in magnitude to estimates in the literature, which range from 100 to 500 hours (French, 2005; French and Jones, 2011; De Bresser, 2024). However, our definition of bad health only includes severe cases that qualify for DI benefits, which may explain the larger estimate. The estimate implies that the marginal utility of leisure is substantially higher for sick individuals than for healthy ones, and that sickness reduces disposable time by roughly 15% $(587/4,000)$.

The relative weight of the bequest motive and its curvature, $\rho_1$ and $\rho_2$, are estimated to be 36.045 and 166,647.485, respectively. While the literature documents substantial heterogeneity in bequest parameters across studies, our estimates are of comparable magnitude to those found in the existing structural literature, which typically finds $\rho_1$ in the range of 0 to 2,000 and $\rho_2$ in the range of 200,000 to 500,000 (French, 2005; De Nardi et al., 2010; French and Jones, 2011; De Bresser, 2024; De Nardi et al., 2025).

Table 4: MSM estimates of preference, health, and bequest parameters

| Parameter | Estimate |
|---|---|
| $\lambda$ | $-0.364$ |
| | (0.002) |
| $\gamma$ | $-0.066$ |
| | (0.019) |
| $\psi$ | 0.041 |
| | (0.001) |
| $\delta$ | 587.174 |
| | (17.044) |
| $\rho_1$ | 36.045 |
| | (1.451) |
| $\rho_2$ | $166,647.485$ |
| | $(5,983.554)$ |

Note: Asymptotic standard errors in parentheses.

Using the structural model estimates, Figures 5 and 6 compare the behavior generated by the model to that observed in the data. The left and right panels present results for the January 1950 and January 1953 cohorts, the two cohorts used in estimation. The model reproduces the targeted empirical moments for labour supply, pension claiming, and asset accumulation remarkably well. It also captures the cross-cohort differences in part-time work: individuals in the 1950 cohort are more likely to choose part-time work without claiming a pension and less likely to combine part-time work with pension benefits, consistent with the empirical patterns.

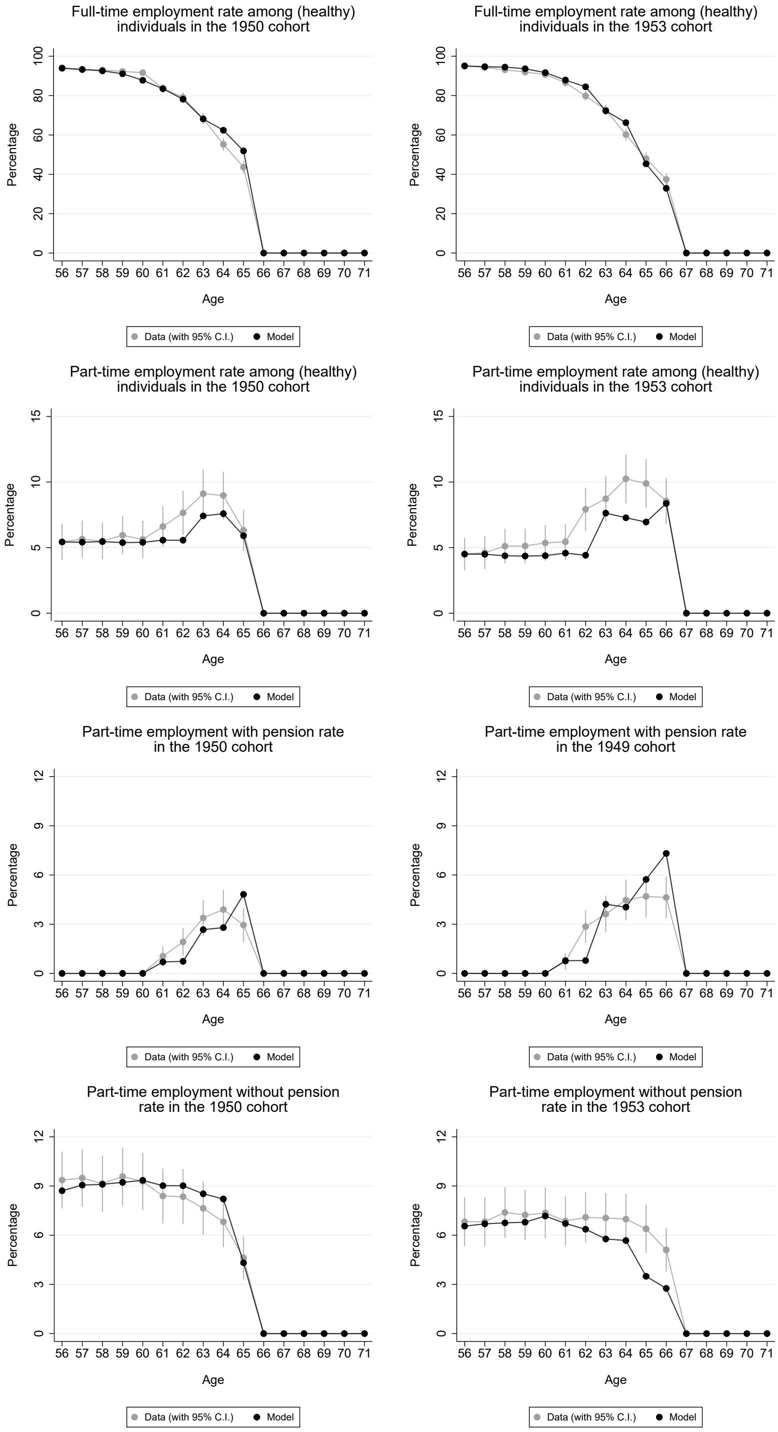



Figure 5: Model fit for the January 1950 and January 1953 estimation cohorts

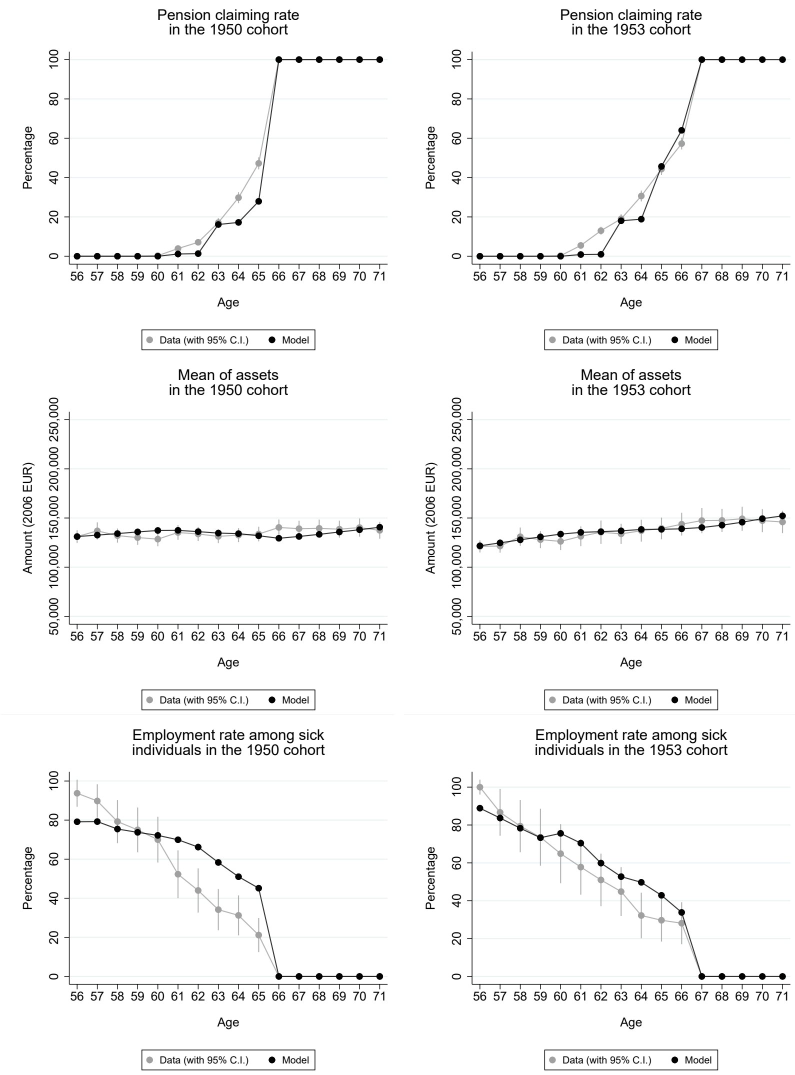



Figure 6: Model fit for the January 1950 and January 1953 estimation cohorts

## 7.2 Model validation

Recent empirical work integrates exogenous policy-induced variation into structural models, strengthening identification and thereby enhancing the credibility of counterfactual policy simulations (Low and Meghir, 2017; Andrews et al., 2020). This approach typically takes one of two forms: researchers either use reforms to estimate the model by fitting its parameters so that it reproduces the behavioral responses induced by the reforms (French et al., 2022; Low and Pistaferri, 2015; Autor et al., 2019; Best et al., 2020; DellaVigna et al., 2017), or they reserve the reform for out-of-sample validation of the model's counterfactual predictions (Todd and Wolpin, 2006; Attanasio et al., 2012; De Bresser, 2024).

The institutional context shaped by two major pension reforms, combined with rich administrative data, allows us to pursue both approaches simultaneously. We first estimate the structural model using the January 1950 and January 1953 cohorts. We then validate the model by applying the estimates reported in Table 4 to simulate life-cycle profiles for the December 1949 cohort and comparing these simulated trajectories to the observed data. This validation exercise is particularly informative because the December 1949 cohort exhibited markedly different retirement patterns relative to those of the January 1950 and January 1953 cohorts (Figure

1). Evaluating whether the model can reproduce these non-targeted differences provides a stringent test of its external validity and strengthens confidence in its ability to generate credible counterfactual policy simulations.

The results are presented in Figures 7 and 8, which show the simulated and observed life-cycle profiles for the December 1949 cohort, the non-targeted cohort used for validation. The figures show that the model reproduces well the labour supply and pension claiming decisions of this cohort. In particular, the model captures the differences across pension regimes: at all ages, the share of individuals claiming a pension is higher for the 1949 cohort than for the January 1950 and January 1953 cohorts, both in the data and in the simulations, while the share working full time is correspondingly lower.

With respect to partial retirement as the focus of our study, the model also replicates well the differences across cohorts in part-time work at older ages. Figure 9 illustrates this by comparing part-time employment rates between the 1949 and 1950 cohorts in the left panel and between the 1950 and 1953 cohorts in the right panel. In both cases, the simulated outcomes align closely with the observed differences induced by the two policy reforms, providing further evidence that the model captures the impact of institutional variation on retirement behavior.

Having established the model's ability to replicate non-targeted patterns, we now turn to policy simulations that are empirically difficult to observe, predicting behavioral responses under counterfactual regimes.

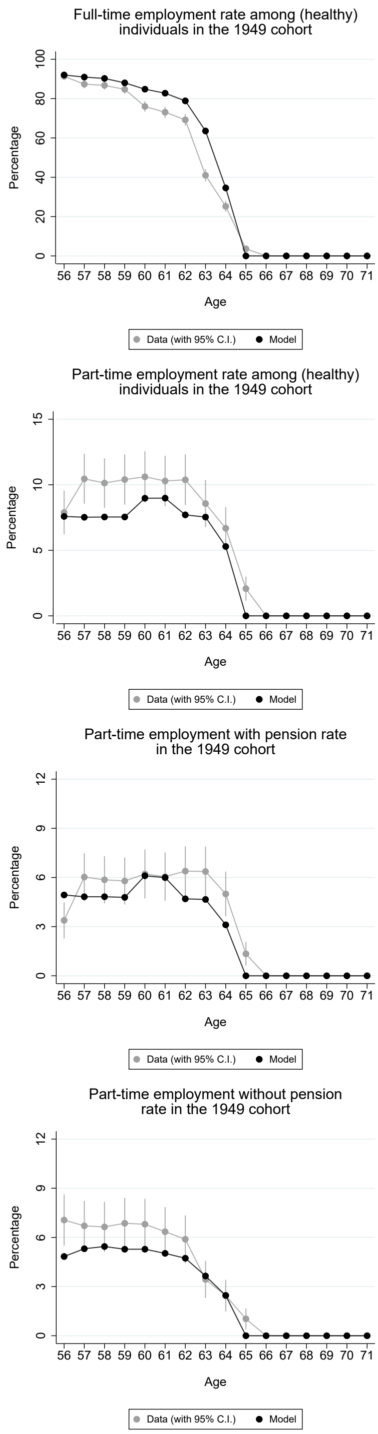



Figure 7: Model fit for the January 1949 out-of-sample cohort

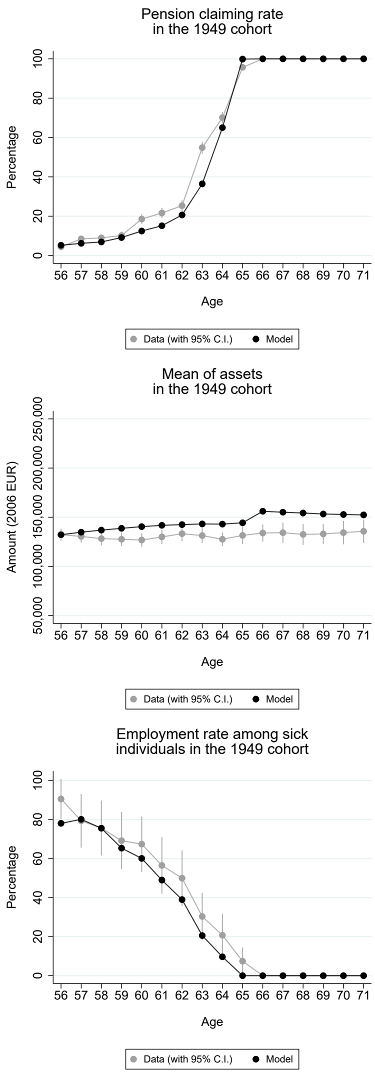



Figure 8: Model fit for the January 1949 out-of-sample cohort

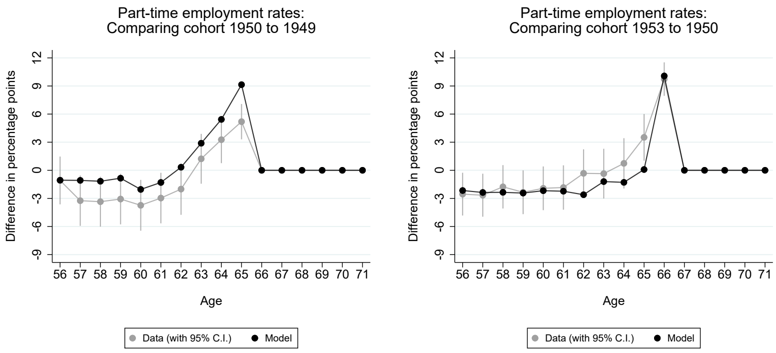



Figure 9: Model fit for part-time work across birth cohorts

# 8 Policy simulations

The descriptive patterns in Figures 2 and 3, together with the causal evidence on the impact of the early-retirement reform in the left panel of Figure 4, suggest that access to partial pensions, DI benefits, or generous pension provisions allows individuals to supplement part-time earnings with benefit income, thereby facilitating consumption smoothing and making partial retirement financially feasible. Building on this insight, we use the structural model to simulate three policy interventions. While all interventions we simulate may, in principle, enhance the affordability of partial retirement, each operates through a different economic mechanism and reshapes the incentive structure of partial retirement in a distinct way. In the first simulation individuals are permitted to claim partial pension benefits while working part-time, compared to a baseline where this option is not available. This expands the set of feasible intertemporal allocations of accrued pension rights. This policy facilitates consumption and leisure smoothing but does not increase lifetime income. By contrast, the second and third simulations introduce direct financial incentives for partial retirees through wage and pension-accrual subsidies. Wage subsidies raise income during the years of part-time work, whereas pension-accrual subsidies increase income during the subsequent phase of full retirement. We conduct each simulation for three cohorts and retain all cohort-specific differences in pension rules generated by the two reforms studied in this paper. For example, the 1953 cohort continues to face a higher SRA than the 1949 and 1950 cohorts. In doing so, we show how a given policy intervention generates heterogeneous behavioral responses under different pension regimes.

## 8.1 Access to partial pension

Since 2006, compulsory occupational pension schemes in the Netherlands have allowed individuals to claim part of their accrued occupational pension rights while working part-time, from the cohort-specific early-retirement age onward. This option has only recently been adopted in several other countries.[7] To assess its implications, we simulate individual behavior in a setting where individuals can combine part-time wages with partial pensions until their SRA, as in the agents' optimization problem described in Section 6.1, and compare this to a benchmark in which this option is removed.

Figure 10 presents the simulation results. Allowing individuals to claim partial occupational pension rights raises part-time employment in all cohorts. For younger cohorts, the increase emerges at later ages. This timing pattern is consistent with institutional eligibility rules: only the 1949 cohort can access occupational pension rights before age 60, whereas the 1953 cohort cannot claim the state pension until age 66 and 4 months (Table 1). At the same time, the effects are stronger for the 1953 cohort. For example, access to partial pensions raises part-time employment by about 4 percentage points between ages 63 and 66, whereas the increases for older cohorts remain below this level at every age. This stronger response reflects that, for the 1953 cohort, partial-pension eligibility relaxes a much tighter liquidity and timing constraint. With a higher SRA, this cohort faces a longer period in which reduced hours would otherwise imply low income, making the partial pension a more valuable income source for smoothing consumption. Unlike the oldest cohort, they were not eligible for generous early-retirement pension provisions.

Notably, the rise in part-time employment is drawn almost equally from declines in full-time employment and in full retirement. This pattern indicates that the policy has only a negligible

[7] OECD (2025) reports that only 11 of the 33 surveyed countries place no restrictions on combining wages and partial pensions before the state pension age. The United States introduced a phased retirement program for federal employees in 2014, which allows employees to work part-time while drawing a partial pension. Among the recent adopters, Switzerland introduced this option in 2024 and Austria in 2026.

effect on total hours worked at any given age.

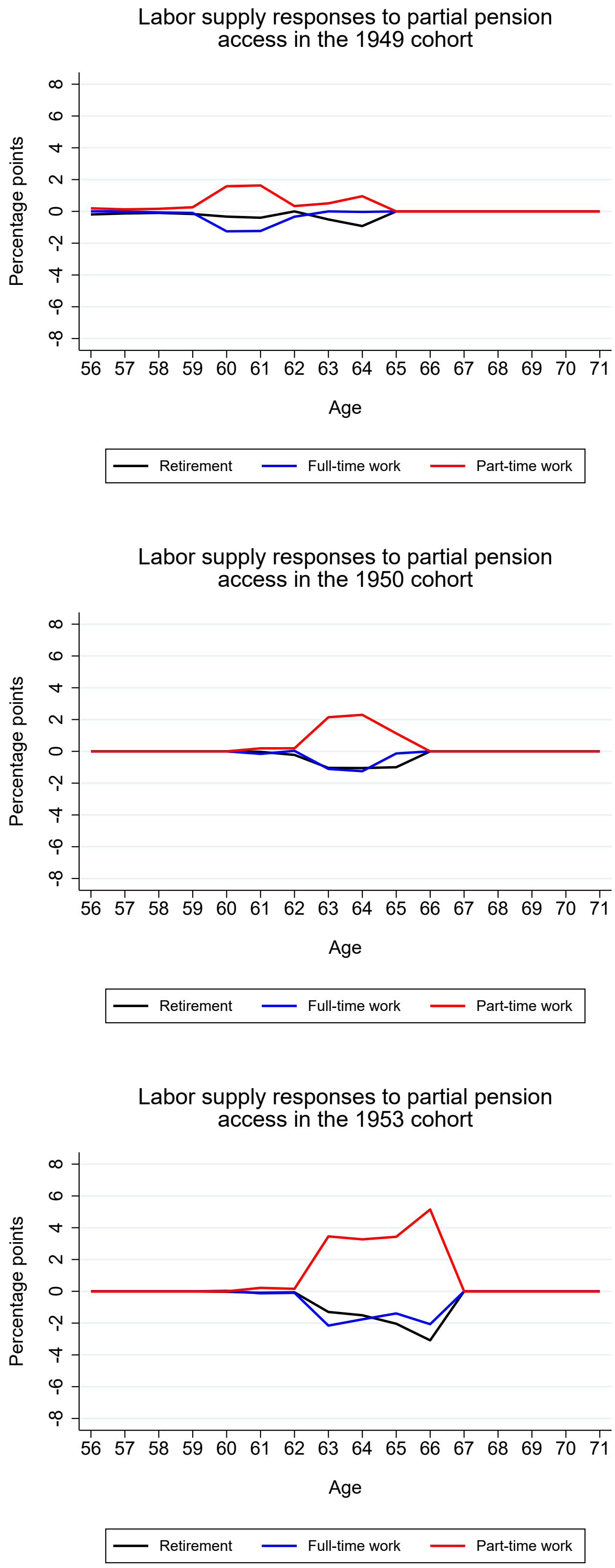



Figure 10: Labor-supply responses to partial-pension access.

## 8.2 Wage subsidy during partial retirement

By the late 2010s, an increasing number of Dutch collective labor agreements introduced subsidized partial-retirement arrangements (Generation Pact). These schemes typically begin at age 60 or 62, depending on the sector, and apply in the years leading up to the SRA. They allow employees to reduce working hours with a less-than-proportional reduction in wages and continued pension accrual based on the full-time wage (Section 2). Therefore, the Generation Pact provides two subsidies: a wage subsidy and a pension-accrual subsidy. Employees may reduce their hours from any initial level, provided that the reduced schedule does not fall below 0.5 FTE. Sectoral agreements often offer multiple options. Two widely used examples are: employees who work 80 percent of their former hours receive 90 percent of their former wage, and employees who work 50 percent of their former hours receive 60 percent of their former wage. These schemes do not provide access to partial pension rights; workers who wish to draw partial pension benefits must file a separate claim with their occupational pension fund.

Here we simulate the effect of subsidizing wages during partial retirement by comparing outcomes to a benchmark without any subsidy. In our structural model, individuals can choose between two partial-retirement schedules: 0.5 FTE or 0.8 FTE (Section 6.1). In the subsidy scenario, employees working 0.5 FTE receive 60% of their former wage, while those working 0.8 FTE receive 90% of their former wage. In the next section, we examine the effect of subsidizing pension accrual, thereby disentangling the respective impacts of the two subsidies. In these simulations, we do not allow access to partial pension rights. Moreover, individuals may begin partial retirement at any age from the cohort-specific early-retirement age onward, rather than only at the commonly observed Generation Pact ages of 60 or 62. Apart from these adjustments, the simulations replicate the institutional features of the Generation Pact described above.

Figure 11 presents the results. Wage subsidies increase part-time employment, but the magnitude of the effect differs substantially across cohorts. For the 1949 cohort, the increase is less than one percentage point, whereas for the 1950 and 1953 cohorts it is considerably larger, reaching between six and eight percentage points. Because partial pension benefits are not available in this simulation, the relevant comparison is between subsidized part-time income and full retirement with cohort-specific pension benefits. For the 1949 cohort, full retirement provides relatively generous pension income, so the wage subsidy adds little to the attractiveness of partial retirement: retiring fully already offers a high and liquid income stream. By contrast, the 1950 and 1953 cohorts face substantially lower pension income upon full retirement. For them, the wage subsidy raises income during partial retirement relative to their full-retirement option, making liquidity constraints less binding and thereby increasing the attractiveness of gradual retirement.

For the 1950 and 1953 cohorts, the sizable increases in partial retirement are matched by declines in both full-time work and retirement, with the drop in full-time employment being particularly pronounced. As a result, total hours worked decline slightly for the 1950 and 1953 cohorts, by roughly $-1\%$ to $-2\%$ between ages 61 and 66, whereas the 1949 cohort shows almost no change.

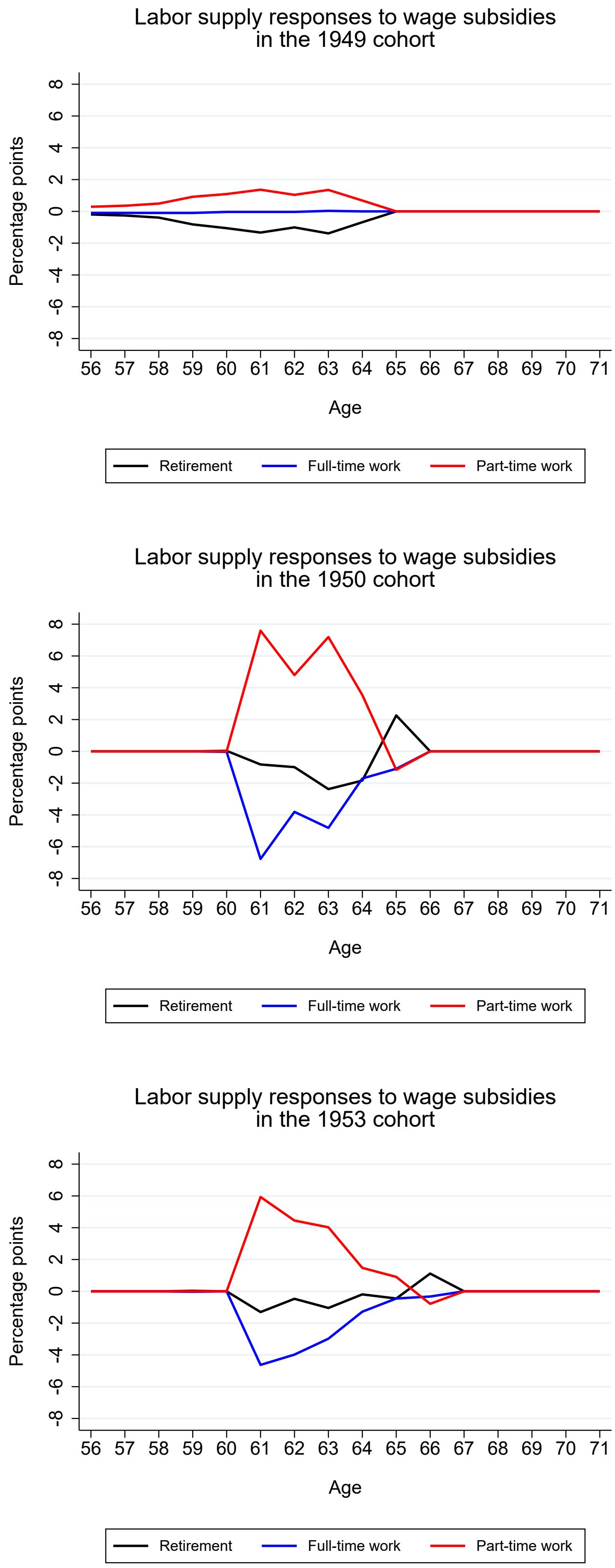
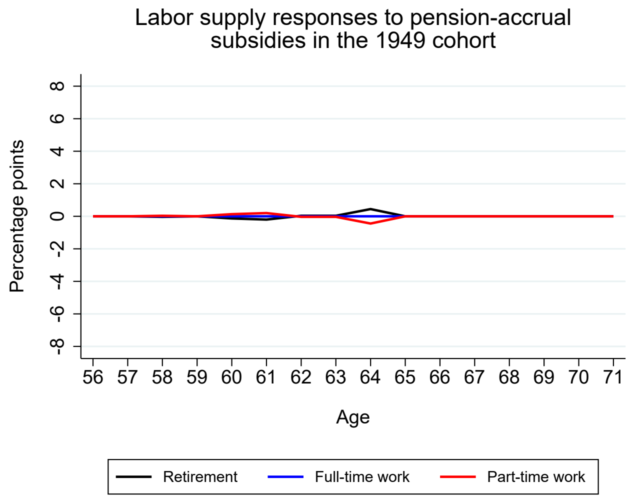
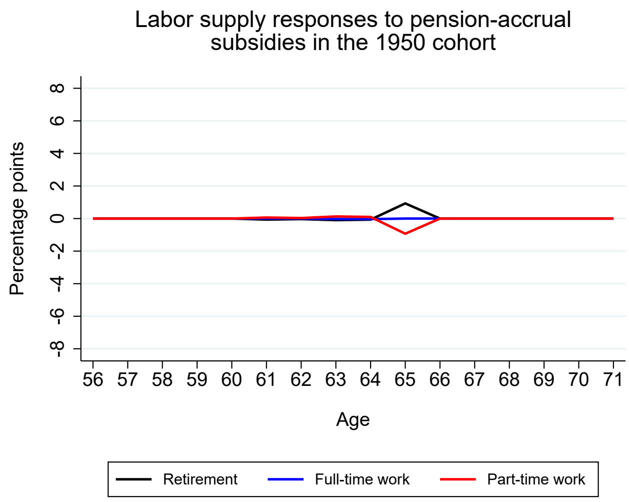



Figure 11: Labour-supply responses to wage subsidies during partial retirement.

### 8.3 Pension-accrual subsidy during partial retirement

Here we simulate the effect of subsidizing pension accrual during partial retirement, as in the Generation Pact, and compare this to a benchmark in which there is no subsidy. Comparing Figure 11 and Figure 12 shows that the two subsidies have sharply different effects on optimal behavior. In the estimated life-cycle model, shifting the subsidy from current wages to future pension accrual leads to a near-zero effect on partial retirement. This is notable because a back-of-the-envelope calculation indicates that the two subsidies raise lifetime income by similar amounts once both are expressed in present-value terms: about €5,000 for the wage subsidy and €6,000 for the pension-accrual subsidy. The behavioral difference arises because the two subsidies differ fundamentally in whether they relax liquidity constraints. A wage subsidy increases disposable income at the moment hours are reduced, relaxing liquidity constraints and allowing workers to smooth consumption during partial retirement. Additional pension accrual instead raises future annuity income that is illiquid, cannot be borrowed against, and is subject to survival uncertainty. Because these features reduce the value of future annuity income relative to current disposable income, workers place a higher value on a euro of wage subsidy than on an actuarially equivalent euro of pension accrual. This finding lends support to our main prediction that policies enabling consumption smoothing make partial retirement financially feasible.



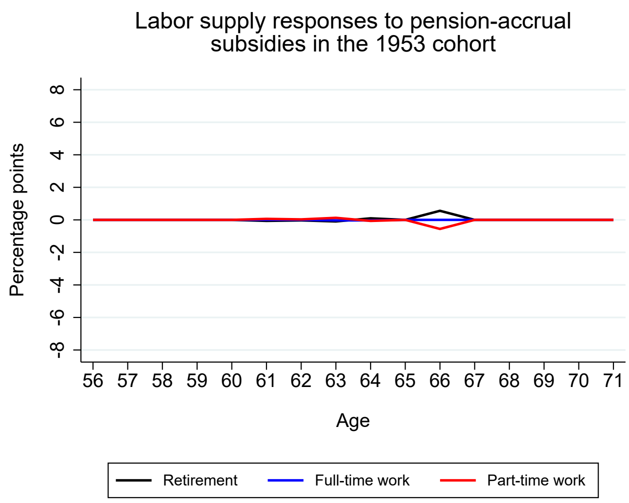




Figure 12: Labour-supply responses to pension-accrual subsidies during partial retirement.

# 9 Conclusion

We study partial retirement behavior using a life-cycle model that explains forward-looking choices over labour supply, pension claiming, and consumption. We exploit two pension reforms: one allows us to estimate the model, and the other to validate its predictions. Our administrative data are exceptionally rich and allow us to observe individuals' pension wealth directly, rather than reconstruct it from earnings histories. They also provide exact timing of hour reductions and pension claims, allowing us to observe transitions that would be undercounted in lower-frequency data. We incorporate the full set of institutional rules so that retirement incentives are modeled realistically.

We address both theoretical and policy questions. First, we provide empirical evidence on a core prediction of the standard life-cycle model. In that model, individuals gradually reduce hours as they age provided they can smooth consumption while working less. In reality, consumption smoothing is not guaranteed. We show that partial retirement occurs precisely when individuals have the income sources needed to maintain consumption during this transition. For example, at age 63, participation in partial retirement increases by about 4 percentage points when individuals are allowed to claim partial pension rights in proportion to their reduction in hours, or alternatively when part-time wages are subsidized by 10 percentage points above proportional pay.

Second, this insight offers a new explanation for the limited prevalence of partial retirement. We argue that when workers lack the financial means to smooth consumption as hours fall, participation in partial retirement arrangements becomes infeasible. Existing studies attribute the scarcity of partial retirement either to nonconvexities in labour supply, such as fixed costs or indivisible hours, or to demand-side constraints that limit access to part-time jobs. Our results point to a different mechanism. Even when the hourly wage for part-time work is lower, our model predicts substantial partial retirement once workers have access to income sources that allow them to smooth consumption as they scale back their hours. This highlights liquidity and consumption smoothing, rather than hourly wage levels or job availability, as the binding constraints.

Third, our findings speak directly to policy design. As retirement ages rise, many workers cannot or do not wish to remain in full-time employment until reaching the new eligibility thresholds. If policymakers aim to make partial retirement a viable alternative, income instruments that facilitate consumption smoothing are effective. The challenge is to structure these instruments so that partial retirement substitutes for full retirement rather than for continued full-time work. Only under those conditions can such policies increase total labour supply.

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# Online Appendix

## A Stylized facts

### A.1 Truncated distribution of full-time equivalent

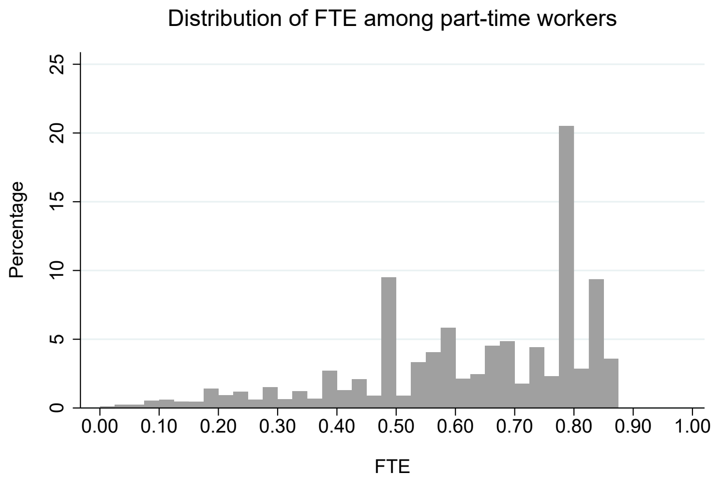


Figure 13: Truncated distribution of full-time equivalent values (FTE $>0$ and FTE $<0.875$).

### A.2 The fraction of accrued pension rights claimed while working part-time

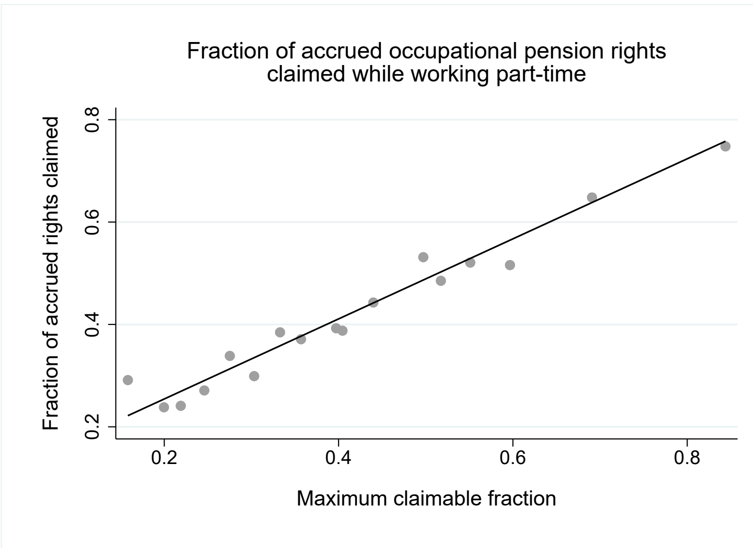


Figure 14: Fraction of accrued occupational pension rights claimed while working part-time.

Figure 14 plots the average fraction of accrued occupational pension rights that individuals claim while working part-time, conditional on claiming any fraction of accrued rights, against the maximum fraction they can claim, given by $1 - \text{FTE}$. The claimed fraction is computed as the ratio of pension rights claimed in the last year of partial retirement to those claimed in the first year of full retirement. The full-time equivalent corresponds to the level of employment in the last year of partial retirement. The figure shows that individuals typically claim the maximum possible amount, which supports our modeling assumption.

### A.3 Fraction of individuals claiming accrued occupational pension rights

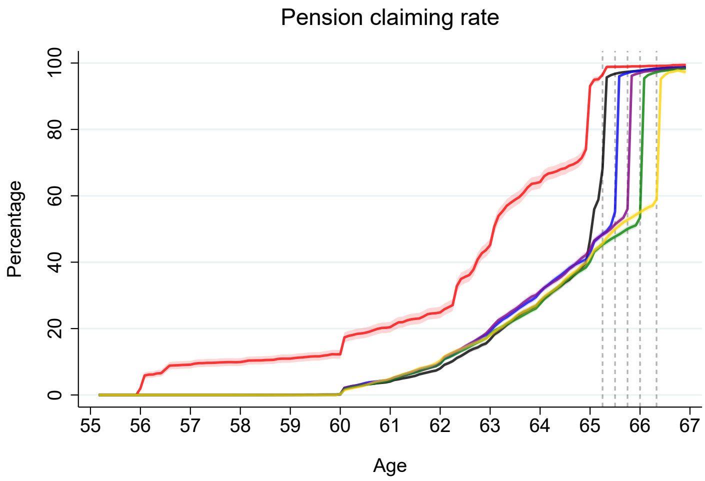


Figure 15: Fraction of individuals claiming accrued occupational pension rights.

Figure 15 plots the fraction of individuals who claim their accrued occupational pension rights. Vertical reference lines indicate the state pension eligibility ages. The figure shows that almost all individuals claim their occupational pension rights as soon as they become eligible for the state pension.

# B Robustness to a regression discontinuity approach

In section 5, the DiD estimates of the effects of the early retirement and SRA reforms rely on the assumption that, in the absence of the reforms, the outcome variable would have followed the same age-related trend in the treatment and control groups. We showed that the treatment and control groups exhibit no systematic differences in the outcome during the pre-treatment period. Here we argue that the fact that we find significant effects of the reforms does not depend on the specific identifying assumption we made. We consider an alternative identification strategy that relies on different identifying assumptions, and the results confirm the results based on the DiD method.

## B.1 RD model

We exploit the birth-date cut-offs that assigned individuals to different pension eligibility rules as a source of exogenous variation in treatment status. We rely on a sharp RD design to estimate the effect of the two reforms. In particular, the discontinuous jumps at the cut-offs identifies the treatment effect of interest which can be formalized as

$$\lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x] \tag{14}$$

where $X_i$ is the birth date and $c$ a cut-off point. For the early retirement reform, the cut-off is January 1, 1950: individuals born in December 1949 or earlier retained access to the generous early-retirement occupational pensions, whereas those born just after the cut-off were no longer eligible (Table 1). We compare individuals born in December 1949 with those born in January 1950, two adjacent birth cohorts, examining their behavior at the statutory retirement age of 65 years and 3 months.

With respect to the SRA reform, individuals born in January 1950 or later were no longer eligible for the generous early-retirement occupational pensions, and successive birth cohorts within this group became subject to progressively higher SRAs. We compare individuals born in December 1952 with those born in January 1953, two adjacent birth cohorts. We examine their behavior at the statutory retirement age of the treated group, which is 66 years and 4 months.

The sharp RD design relies on two main assumptions. The first requires a discontinuous jump in treatment status at the cut-off. This condition is satisfied by construction in our setting: all individuals $i$ with $X_i \geq c$ are assigned to the treatment group, while those with $X_i < c$ belong to the control group.

The second assumption requires continuity in potential outcomes as a function of the assignment variable around the cut-off. This implies that, in the absence of the reform, the outcome variable would not exhibit a discontinuous jump at the cut-off. In other words, all other factors affecting the outcome must evolve smoothly at the threshold. Although this assumption cannot be tested directly, one can examine whether relevant covariates exhibit significant changes at the cut-off. Since we compare individuals born only one month apart, we assume that these covariates do not vary meaningfully at the cut-offs.

## B.2 Reduced-from estimates

Figure 16 presents the RD plots. Each plot shows local linear fits for an outcome using a symmetric bandwidth of thirty days around the cut-off date. The outcomes are working full-time and working part-time. The left panel corresponds to the early retirement reform, and the right panel to the SRA reform. Vertical lines indicate the birth dates at which different pension rules apply. The plots show clear discontinuities at the cut-off points, in the expected directions. Moreover, the magnitudes of the jumps are consistent with the DiD estimates shown in Figure 4. Overall, both identification strategies provide evidence that the early retirement and SRA reforms significantly altered incentives to work full-time and part-time.

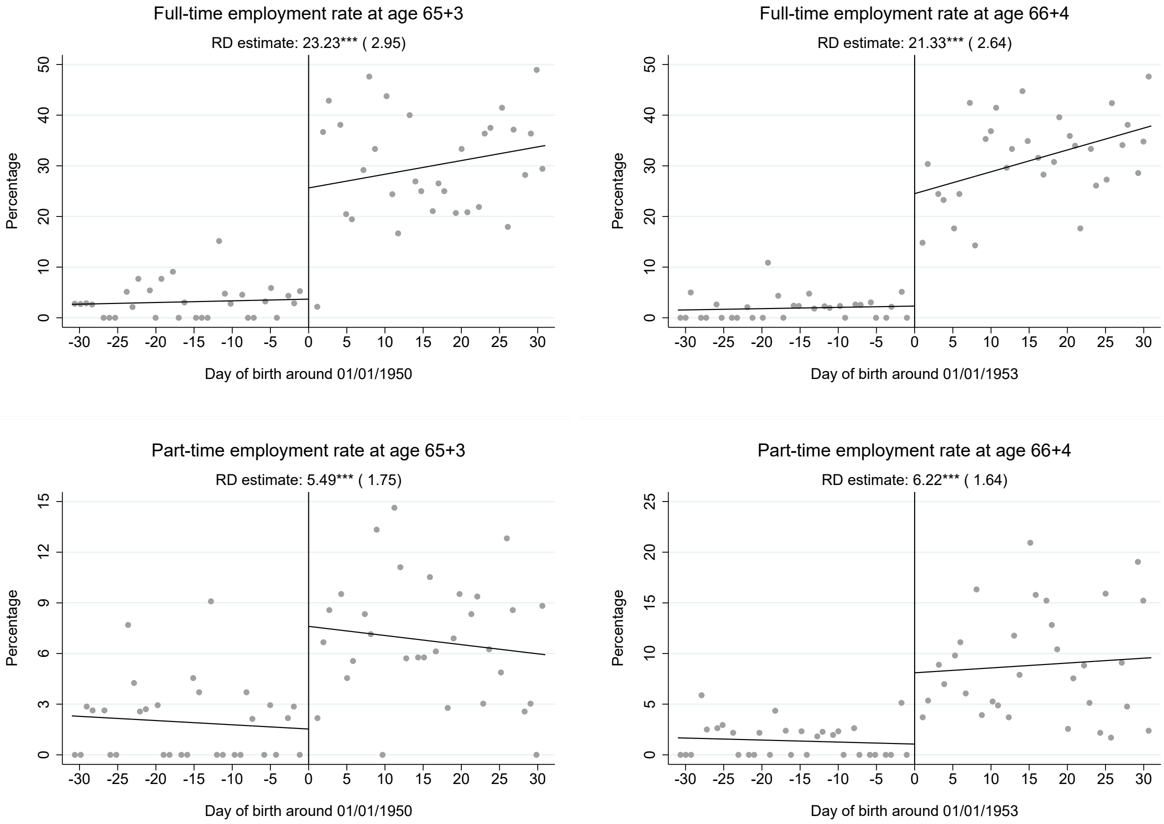


Figure 16: Impact of the early retirement (left panel) and SRA (right panel) reforms.

# C Details on the model setup

## C.1 Model notation

Table 5: Notation for model variables and feasible sets

| Symbol | Description |
|---|---|
| $d_{it}$ | Vector of decision variables |
| $\mathcal{D}(x_{it})$ | Feasible decision set given state $x_{it}$ |
| $h_{it}$ | Annual hours of work (decision variable) |
| $\mathcal{H}$ | Baseline choice set for $h_{it}$ |
| $o_{it}$ | Occupational pension claiming (binary decision) |
| $\mathcal{O}$ | Baseline choice set for $o_{it}$ |
| $c_{it}$ | Consumption (decision variable) |
| $\mathcal{C}(h_{it}, o_{it}, x_{it})$ | Feasible consumption set given $(h_{it}, o_{it}, x_{it})$ |
| $x_{it}$ | Vector of state variables |
| $t$ | Age in years (state variable) |
| $w_{it}$ | Annual full-time wage (state variable) |
| $p_{it}$ | Accrued occupational pension rights (state variable) |
| $v(d_{it}, x_{it})$ | Occupational pension actuarial adjustment factor |
| $e_{it}$ | Occupational pension annuity entitlement |
| $g_{it}$ | State pension annuity entitlement (AOW) |
| $g(k_i)$ | State pension offset |
| $t^g$ | State pension eligibility age |
| $s_{it}$ | Retirement status (state variable) |
| $k_i$ | Birth cohort (state variable) |
| $a_{it}$ | Assets (state variable) |
| $\mathcal{A}(x_{it})$ | Feasible asset set given state $x_{it}$ |
| $r$ | Constant net return on assets |
| $m_{it}$ | Health status (state variable) |
| $l_{it}$ | Leisure as a function of $h_{it}$ and $m_{it}$ |
| $q_{it}$ | Bequest motive as function of $a_{it}$ |
| $y_{it}$ | Earnings from wages |
| $f_{it}$ | Full-time equivalence |
| $\eta$ | Part-time wage multiplier |
| $b_{it}$ | Disability insurance benefit |
| $\pi_t$ | Conditional survival probability in period $t$ |
| $I_{it}$ | Taxable income |
| $T_{it}$ | Income tax liability |
| $A_{it}$ | Asset tax liability |
| $L_{it}$ | Labor tax credit |
| $\tau_{it}$ | Net income |

Table 6: Notation for model parameters

| Symbol | Description |
|---|---|
| $\lambda$ | Curvature of utility with respect to consumption $c_{it}$ |
| $\gamma$ | Curvature of utility with respect to leisure $l_{it}$ |
| $\psi$ | Relative weight of leisure in utility |
| $\delta$ | Health time penalty parameter |
| $\rho_1$ | Bequest motive strength parameter |
| $\rho_2$ | Bequest shift parameter |
| $\beta$ | Time preference parameter |
| $z(k_i)$ | Occupational pension accrual rate parameter |
| $\alpha$ | Intercept in wage process |
| $\phi$ | Autoregressive coefficient for wage process |
| $\epsilon_{it}$ | Wage shock (innovation term) |
| $\sigma_{\epsilon_{it}}$ | Standard deviation of wage shock |
| $\mu_0$ | Baseline health risk parameter |
| $\mu_1$ | Age effect on health risk |
| $\mu_2$ | Health persistence parameter |
| $\kappa^j$ | Contribution rate for scheme $j \in \{\mathrm{WW}, \mathrm{ZVW}, \mathrm{VV}, \mathrm{ABP}\}$ |

## C.2 Social insurance and tax schedules

We use the institutional rules and parameter values for social security and tax schedules that were in force in 2006. We take 2006 as the reference year because this is the first year in which we observe all variables needed to initialize the agents' decision problem (Section 3). Moreover, although later years may feature different rules or updated nominal amounts due to routine indexation, we keep the 2006 schedules fixed. Allowing year-specific rules would require embedding time-varying institutional rules and parameter values directly into the dynamic programming problem, substantially increasing the complexity of both the model and its estimation. For example, in 2006 employees paid unemployment insurance (WW) contributions, while employers paid their own WW contributions separately. As part of payroll tax reforms implemented in 2009, the employee WW contribution was abolished and WW financing has since been borne entirely by employers. Routine inflation adjustments do not affect our modeling because we convert all nominal amounts into 2006 euros using the consumer price index published by Statistics Netherlands.

### Social security contributions

Social security contributions are employee-insurance premiums that finance earnings-related social insurance schemes. They are tied to employment and are typically paid by employers, and historically partly by employees. Until 2006, employees contributed to the unemployment insurance scheme under the Dutch Unemployment Act (Werkloosheidswet, WW), with contributions assessed on earnings according to

$$\kappa_{it}^{WW} = \begin{cases} 0, & y_{it} + b_{it} < 15{,}562, \\ 0.058\left(y_{it} + b_{it} - 15{,}562\right), & 15{,}562 \leq y_{it} + b_{it} < 44{,}800, \\ 0.058\left(44{,}800 - 15{,}562\right), & y_{it} + b_{it} \geq 44{,}800, \end{cases}$$

where $y_{it}$ denotes annual gross earnings and $b_{it}$ denotes DI benefits, as defined in Section 6.

Employees are also required to contribute to the universal public health insurance scheme under the Dutch Health Insurance Act (Zorgverzekeringswet, ZVW). The employee contribution is given by

$$\kappa_{it}^{ZVW} = \begin{cases} 0.0125\,(y_{it} + b_{it}) + 399, & y_{it} + b_{it} < 30{,}320, \\ 0.0125 \cdot 30{,}320 + 399, & 30{,}320 \leq y_{it} + b_{it} < 33{,}514, \\ 0, & y_{it} + b_{it} \geq 33{,}514. \end{cases}$$

### Taxable income

Social security contributions to WW and ZVW are deducted from gross income to obtain taxable income:

$$I_{it} = y_{it} + b_{it} + e_{it} + g_{it} - \kappa_{it}^{WW} - \kappa_{it}^{ZVW},$$

where $g_{it}$ denotes state pension annuity entitlement.

### National insurance contributions

National insurance contributions (premies volksverzekeringen, VV) are universal social insurance contributions that finance population-wide programs that are not tied to employment.

Every resident pays these contributions through the income tax system. They cover the state pension (AOW), surviving dependants' benefits (ANW), and long-term care (WLZ). VV contributions are assessed on taxable income in the lower part of the income distribution and are paid by individuals rather than employers. They apply only to the first two taxable income brackets and depend on whether the individual has reached the statutory state pension age $t^g$. Formally, the liability is defined as

$$\kappa_{it}^{VV} = \begin{cases} 0.324\, I_{it}, & I_{it} < 30{,}371 \text{ and } t < t^g, \\ 0.324 \cdot 30{,}371, & I_{it} \geq 30{,}371 \text{ and } t < t^g, \\ 0.145\, I_{it}, & I_{it} < 30{,}371 \text{ and } t \geq t^g, \\ 0.145 \cdot 30{,}371, & I_{it} \geq 30{,}371 \text{ and } t \geq t^g. \end{cases}$$

### Income tax

Income tax is levied at the individual level, meaning that each person's earnings are taxed separately rather than on a household basis (OECD, 2004). The tax base is taxable income $I_{it}$, which includes DI benefits. The income tax liability is given by

$$T_{it} = \begin{cases} 0.01\, I_{it}, & I_{it} < 16{,}721, \\ 0.01 \cdot 16{,}721 + 0.0795\,(I_{it} - 16{,}721), & 16{,}721 \leq I_{it} < 30{,}371, \\ 0.01 \cdot 16{,}721 + 0.0795\,(30{,}371 - 16{,}721) \\ \quad +0.42\,(I_{it} - 30{,}371), & 30{,}371 \leq I_{it} < 52{,}072, \\ 0.01 \cdot 16{,}721 + 0.0795\,(30{,}371 - 16{,}721) \\ \quad +0.42\,(52{,}072 - 30{,}371) \\ \quad +0.52\,(I_{it} - 52{,}072), & I_{it} \geq 52{,}072. \end{cases}$$

### Asset tax

Prior to 2017, the Dutch income tax system included a levy on net assets. Taxpayers were assumed to earn a fixed return of 4% on net assets above the statutory threshold, which was then taxed at a rate of 30%. This implied an effective tax burden of 1.2% on assets exceeding €39,568. Formally, the liability is specified as

$$A_{it} = \begin{cases} 0, & a_{it} < 39{,}568, \\ 0.012\,(a_{it} - 39{,}568), & a_{it} \geq 39{,}568, \end{cases}$$

where $a_{it}$ denotes the asset balance of individual $i$ in period $t$.

### Tax credits

Taxpayers are entitled to two major tax credits. First, the general tax credit (algemene heffingskorting) applies and amounts to €1,876. It is deducted directly from the calculated income tax liability. Second, the labour tax credit (arbeidskorting) is deducted from income tax and depends on the level of annual earnings $y_{it}$, according to

$$L_{it} = \begin{cases} 0.01753\, y_{it}, & y_{it} < 8{,}328, \\ 0.01753 \cdot 8{,}328 \;+\; 0.11213\,(y_{it} - 8{,}328), & 8{,}328 \leq y_{it} < 18{,}147, \\ 0.01753 \cdot 8{,}328 \;+\; 0.11213\,(18{,}147 - 8{,}328), & y_{it} \geq 18{,}147. \end{cases}$$

### Occupational pension contributions

Employees contribute to the occupational pension fund ABP on gross wages above the statutory state pension offset $g(k_i)$. The individual contribution is defined as

$$\kappa_{it}^{ABP} = \begin{cases} 0, & w_{it} < 9{,}839, \\ 0.0636\, f_{it}\,(w_{it} - 9{,}839), & w_{it} \geq 9{,}839. \end{cases}$$

### Income after contributions and taxes

Net income is given by

$$\tau_{it} = y_{it} + b_{it} + e_{it} + g_{it} - \kappa_{it}^{WW} - \kappa_{it}^{ZVW} - \kappa_{it}^{VV} - \kappa_{it}^{ABP} - T_{it} - W_{it} + 1,876 + L_{it}$$

# D Details on the estimation of the dynamic optimization model

Here we provide additional details on the estimation of the structural model. As with the social insurance and tax schedules discussed in Section C.2, we rely on the institutional rules and parameter values that were in force in 2006 and express all monetary amounts in 2006 prices.

## D.1 Estimation of the wage process

Equation (5) presents the assumed wage process. Each period of the life-cycle model corresponds to an age, and therefore we use annual wage observations to estimate the process. Specifically, we use the wage measured in the individual's birth month. Since wages are set by collective labour agreements and do not vary across birth cohorts in terms of wage-setting rules, we estimate the wage process by pooling birth cohorts. As in our DiD analysis in Section 5, we consider three cohorts: those born in December 1949, January 1950, and January 1953. We do not adjust for the fact that wages are observed only for employed individuals, as the same wage scale applies to all workers regardless of employment status. However, we restrict the estimation sample to observations corresponding to full-time work to avoid any contamination arising from wage changes associated with transitions into part-time employment. We estimate the parameters of the linear wage regression in equation (5) using ordinary least squares. Table 7 presents the coefficient estimates. The autoregressive parameter $\phi$ is close to one, consistent with the estimate in De Bresser (2024). The variance of the error term $\epsilon_{it}$ is estimated to be 0.005.

Table 7: Linear regression model explaining the wage process

| Parameter | Estimate |
|---|---|
| $\alpha$ | 0.28*** |
| | (0.04) |
| $\phi$ | 0.97*** |
| | (0.00) |
| N = 33,698 | |

Standard errors (in parentheses) are clustered at the individual level.

## D.2 Estimation of the health process

Equation (7) presents the assumed health status process. The outcome is a binary indicator of bad health. In our life-cycle model, bad health implies receiving DI benefits. We therefore estimate the health process using information from Statistics Netherlands on DI recipiency. Since disability insurance coverage extends only until the SRA, we use observations up to age 65 for the 1949 and 1950 cohorts and up to age 66 for the 1953 cohort (Table 1). As with wages, we use annual observations from all three birth cohorts. We estimate the parameters of the binary logit model in equation (7) using maximum likelihood. Table 8 presents the estimation results.

Table 8: Binary logit model explaining the health process

| Parameter | Estimate |
|---|---|
| $\mu_0$ | $-5.42^{***}$ |
| | (0.16) |
| $\mu_1$ | 0.01 |
| | (0.02) |
| $\mu_2$ | $10.29^{***}$ |
| | (0.27) |
| N = 43,647 | |

Standard errors (in parentheses) are clustered at the individual level.

### D.3 MSM estimator

We describe the MSM estimator following French and Jones (2011). The objective of MSM estimation is to find the preference vector that generates simulated life-cycle decision profiles that best match their empirical counterparts, as measured by a GMM-style criterion function. The estimator solves

$$\hat{\theta} = \arg\min_{\theta} \varphi(\theta, X)^\top \widehat{W} \varphi(\theta, X),$$

where $\theta$ is an $l \times 1$ vector of unknown parameters. The function $\varphi(\cdot)$ denotes the $k \times 1$ vector of moment conditions, with the $k$-th element defined as $m_k^d(x) - m_k^s(\theta)$, where $m_k^d$ is the moment computed from the data and $m_k^s(\theta)$ is the corresponding moment generated by the model simulation. The matrix $\widehat{W}$ is a $k \times k$ weighting matrix.

Although the optimal weighting matrix is asymptotically efficient, it can perform poorly in finite samples (see, e.g., Altonji and Segal, 1996). For this reason, we use a diagonal weighting matrix whose entries are the inverses of the estimated variances of the empirical moments. For example, the first diagonal element is $\widehat{W}_{1,1} = [\widehat{\mathrm{Var}}(m_1^d)]^{-1}$.

Under the regularity conditions (Pakes and Pollard, 1989; Duffie and Singleton, 1993), the MSM estimator $\hat{\theta}$ is both consistent and asymptotically normally distributed,

$$\sqrt{n}(\hat{\theta} - \theta_0) \ \to_d \ N(0, V),$$

with asymptotic variance–covariance matrix

$$V = \left(1 + \frac{1}{N_{sim}}\right) (D^\top W D)^{-1} D^\top W S W D (D^\top W D)^{-1}.$$

We use three simulation draws per individual in the estimation procedure.[8] The matrix $S$ is the $k \times k$ variance–covariance matrix of the empirical moments, and $D$ is the $k \times l$ Jacobian of the moment vector evaluated at the MSM estimate $\hat{\theta}$,

$$D = \left.\frac{\partial \varphi(\theta, X)}{\partial \theta^\top}\right|_{\theta = \hat{\theta}}.$$

[8] Since the estimator is consistent for a fixed number of simulations (Adda and Cooper, 2003), and because we are not primarily concerned with inference on the parameter estimates, we simulate each individual only three times to reduce computational burden.

We compute $D$ by numerical approximation and estimate $S$ using the bootstrap. Specifically, we use 500 bootstrap samples drawn with replacement, following De Bresser (2024).

### D.4 Asset data preparation

As discussed in Section 6.2, we use the MSM to estimate preferences. The estimator targets empirical moments related to labour supply, pension claiming, and assets for each cohort and age. To construct the asset profiles used in these moments, we perform several adjustments to the raw administrative data.

Since the model features annual decision-making, we match annual life-cycle profiles. Labour supply and pension claiming are measured annually on birthdays. Asset data, however, are obtained from the administrative records of Statistics Netherlands and are observed only on January 1 each year. Therefore, for each individual, we use the asset observation closest to their birthday to construct average asset holdings by age. Administrative asset data are recorded at the household level. Therefore, for married men, we divide reported amounts by the square root of two, following the OECD square-root equivalence scale (OECD, 2018).

Next, we deflate all asset values using the Consumer Price Index published by Statistics Netherlands and express monetary amounts in 2006 euros, consistent with the baseline year used for wages. Finally, business cycle and financial market fluctuations may differentially affect assets across cohorts, while our model does not capture these features. To address this, we net out year effects using a regression approach. Specifically, we estimate a model on a larger panel containing all available cohorts from 1919 to 1956, regressing assets on a full set of age and calendar-year dummies, and subtract the estimated year effects from observed assets.

The resulting adjusted data are used to construct both the asset-related moments (average assets at each age, separately by cohort) and the initial asset holdings at the start of the model ($a_{i,t_0}$).

### D.5 Robustness to reversing the roles of pension reforms used for estimation and validation

In Section 7.1, we exploited the SRA reform in 2011 to estimate the structural model parameters, and the early retirement reform in 2006 to validate these estimates. That is, our method of simulated moments estimates were obtained by selecting the life-cycle decision profiles of the 1950 and 1953 cohorts that best match the empirical moments, exploiting the differences in retirement ages induced by the SRA reform. For out-of-sample validation, we assessed how well the estimated model reproduces the observed retirement behavior of the 1949 birth cohort. We selected this cohort because the changes in pension rules and their behavioral effects were more pronounced under the early-retirement reform, providing a more demanding test of the model's ability to replicate non-targeted pension regimes.

Here we analyze how the model estimates change when we reverse the roles of the two reforms, that is, when we target the 1949 and 1950 cohorts for estimation.[9] Table 9 reports the parameter estimates which are very similar to baseline estimates in Table 4. Consequently, the model fit is nearly identical to the baseline for all three cohorts (results available upon request). Taken together, these findings do not indicate meaningful misspecification.

[9] The number of targeted moments increases to 120, up from 115. Here we have 65 moments for cohort 1949, 55 for cohort 1950, and 60 for cohort 1953. This change reflects two adjustments. First, cohort 1949 can start claiming five years earlier than cohort 1953, while cohort 1953 can claim for one additional year. This expands the set of moments for claiming and for part-time work while claiming, adding eight moments in total. Second, cohort 1949 can work one year less than cohort 1953, which reduces the number of moments for full-time work, part-time work, and work while sick by three.

Table 9: MSM estimates of preference, health, and bequest parameters under alternative reform targets

| Parameter | Estimate |
|---|---|
| $\lambda$ | $-0.364$ |
| | (0.001) |
| $\gamma$ | $-0.066$ |
| | (0.009) |
| $\psi$ | 0.042 |
| | (0.000) |
| $\delta$ | 757.732 |
| | (33.573) |
| $\rho_1$ | 31.072 |
| | (0.955) |
| $\rho_2$ | $138,390.183$ |
| | (901.584) |

Notes: 1. Baseline MSM estimates (Section 7.1) are obtained by targeting cohorts 1950 and 1953, corresponding to the SRA reform. Robustness estimates target cohorts 1949 and 1950, corresponding to the early retirement reform. 2. Asymptotic standard errors in parentheses.